



Application Descriptions

7

Lighting

20

Lighting Sensors

1

Summary

This document specifies the Functional Blocks for sensors in the Lighting Application Domain.

Version 01.07.01 is an Approved Standard.

Document updates

Version	Date	Modifications
1.0	2006.06.08	Draft Proposal
1.1	2007.02.02	- Inclusion of Document References. - Preparation of the Draft Proposal.
1.2	2007.03.20	- Publication of the Approved Standard.
1.3	2008.12.09	- AN080 “FBs for new channels 2005.02” integrated.
1.3	2009.06.26	Editorial update in view of inclusion in the KNX Specifications v2.0.
1.4.01	2010.04.14	- Removed references to legacy papers. - Renamed “Switching sensor Basic” to “Light Switching Sensor Basic”. - AN124 “Photometric Sensors” integrated.
1.4.02	2011.12.14	- Editorial update.
01.04.03	2013.09.06	- AN150 “FB Profiles for existing FBs” integrated. - AN119 “FB MDL” integrated.
01.04.04	2013.10.29	Editorial updates for the publication of KNX Specifications 2.1.
WDv01	2014.09.30	Inclusion of - FB Colour Temperature Setting Sensor (Draft Proposal) - FB Colour Control Sensor (Draft Proposal)
WDV02	2014.10.01	Editorial changes
WDV03	2014.10.06	- Editorial changes - Preparation for WGI Editing meeting 2014, 08th to 09th
WDV04	2014.10.09	Result of WGI Edition meeting (participants: Schneider, Insta)
WDV05	2014.10.24	Editorial Changes
WDv06	2014.11.26	Review of the document (meeting KNXA / Insta) in Lüdenscheld - Editorial Change: Datapoint Relative Set Value Control renamed in Relative Set Value Control Colour Temperature (RSCCT) - Splitting of FB Colour Control Sensor Basic into separate FBs: - Colour Sensor RGB - Colour Sensor xyY
WDv07	2014.12.04	- Editorial Changes - First stable WD for discussion in KNX WGI
WDv08	2015.02.03	- Editorial Changes (Figures of Application Models) - Integration of the motivations for the three different colour sensor FBs
WDv09	2015.02.26	- Changing respective to the WGI meeting 2015022625
WDv10	2015.09.17	- Changing respective for the WGI meeting 2015092223
WDv11	2016.04.27	Further modifications as Input for the next WGI Editing Meeting 20160511
WDv12	2016.05.11	Further modifications as Output of the WGI Editing Meeting 20160511
WDv13	2016.05.18	Final editorial review in preparation of WGI review.
WDv13	2016.07.13	Final version for RfV as result of of WGI review in Online Meeting 20160708: DPT 7.004 should be used for transition times.
RfVv14	2016.07.29	Readying for release for voting
RfVv15	2016.10.06	Indirect Integration of the comments of RfV (WGI meeting from 05 th of October 2016) Definition of new DPT 242.600 has been corrected (in the same way as in document 07_20_02 Lighting Actuators with CC RfVv10)
01.05.01	2016.11.19	Preparation of the Draft for Voting.
01.06.01	2017.10.11	- Proposal for resolution of comments.
01.07.01	2019.10.28	- Correction of the encoding of DPT_Colour_RGBW (251.600). - “Overview of Interface Objects and Properties” moved to Chapter 7/20/9. - Creation of the Approved Standard.

References

- [01] Chapter 3/7/2 "Datapoint Types"
- [02] Chapter 6/30/1 "Runtime Profiles"
- [03] Chapter 7/10/1 "HVAC Sensor Functional Blocks"
- [04] Chapter 7/10/2 "HVAC HMI Functional Blocks"
- [05] Chapter 7/10/3 "HVAC Actuator Functional Blocks"
- [06] "Principles Governing Photometry", BIPM, 1983, Monographie 1983-1

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Abbreviations

Datapoints

ASC	Absolute Setvalue Control
ACTC	Absolute Colour Temperature Control
CCT	Current Colour Temperature
CC_RGB	Current Colour_RGB
CC_xyY	Current Colour_xyY
BE	Brightness External
BT	Brightness Threshold
CS_RGB	Colour Set_RGB
CS_RGBW	Colour Set_RGBW
RC_RGB	Colour Relative Control RGB
RC_RGBW	Colour Relative Control RGBW
CS_xyY	Colour Set_xyY
RC_xyY	Colour Relative_xyY
CT_xyY	Colour Transition_xyY
IL	Indoor Luminance
IOO	Info OnOff
MT	Movement Trigger
RI	Room Illumination
RBRCT	Relative Brightness_Colour Temperature Control
RSC	Relative Setvalue Control
RSCCT	Relative Setvalue Control Colour Temperature
SOO	Switch On Off
TSS	Timed StartStop

Parameters

RGB_CCV	Absolute RGB Colour Control Value
XYX_CCV	Absolute xyY Colour Control Value
XYX_CCTT	Absolute xyY Colour Control Transition Time
ACT_CV	Absolute Colour Temperature Control Value
ACT_TT	Absolute Colour Temperature Transition Time
AS	Absolute Setvalue
EBI	Enable Brightness Independency
ETM	Enable Toggle Mode
MSLT	Movement Signal Lowpass Timer
MSPT	Movement Sensor Pause Time
MTE	Movement Trigger Evaluation
OCT	Output Control Timer
OOA	On Off Action
RT	Repetition Time
TC	Trigger Continue

Other

COV	Change Of Value
DSB	Dimming Sensor Basic
HMI	Human Machine Interface
IBS	Indoor Brightness Sensor
ILS	Indoor Luminance Sensor
LSAB	Light Switching Actuator Basic
LSSB	Light Switching Sensor Basic
MDL	Movement Detector for Lighting
CTSS	Colour Temperature Setting Sensor
CTA	Colour Temperature Actuator
CSSRGB	Colour Setting Sensor RGB

CARGB	Colour Actuator RGB
CSSXYY	Colour Setting Sensor XYY
CAXYY	Colour Actuator XYY

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1 Sensors for human interaction

1.1 FB Light Switching Sensor Basic (Object Type 421)

1.1.1 Aims and objectives

The FB Light Switching Sensor Basic is used in the Application Domain Lighting for providing input data to switching actuators. It specifies the functionality, for example contained in a switch or a push button, to switch the switching actuator on or off.

Display elements (with own FBs) can be integrated to show the status of the switching actuator.

The inputs and outputs of the Functional Block are described but not the Human Machine Interface (HMI). Consequently, the manufacturers of the button or switch have the possibility to implement their design and their operation methods.

1.1.2 Functional specification

1.1.2.1 Overview

The FB Light Switching Sensor Basic provides hardwired inputs for triggering transmission of values of output Datapoints.

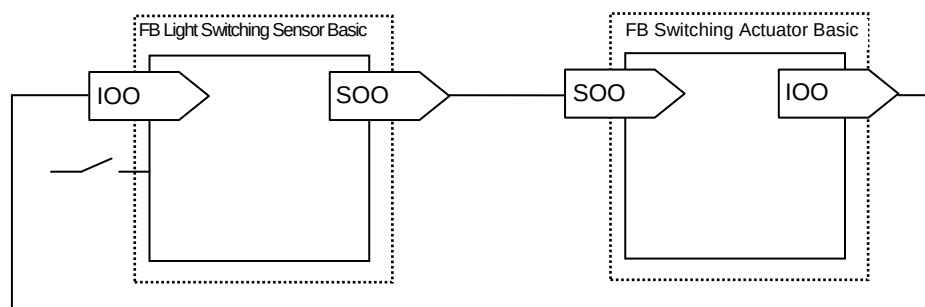
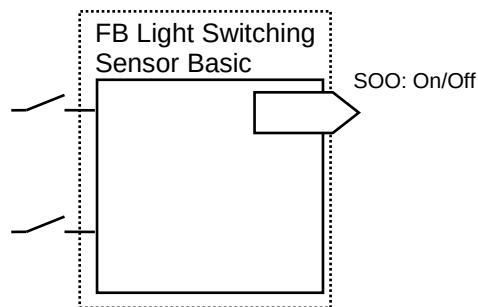


Figure 1 – Basic communication model (example)

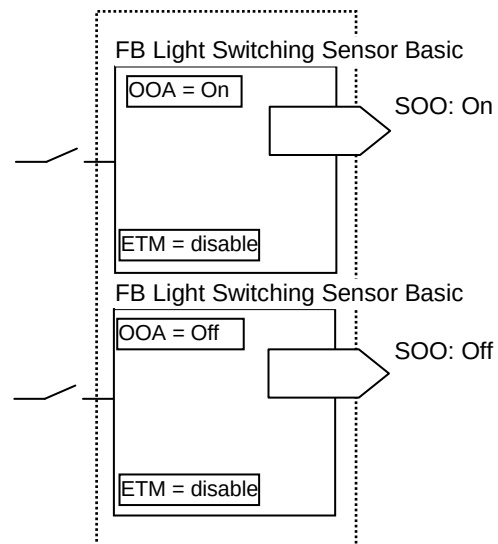
1.1.2.2 Parameter On Off Action

The parameter On Off Action shall limit the values transmitted by the output SOO to either only 0 (“Off”) or 1 (“On”).

This mainly makes sense if an appliance is realised as a combination of two FBs LSSB each with one interaction point. This realisation however is only meaningful if the parameter ETM is not implemented or has the value “Disable”.



**Figure 2 – One FB
with two interaction points**



**Figure 3 – Two FBs
each with one interaction points**

The dashed outlined areas realise the same functionality. The DP SOO in the left solution sends both On and Off. The DPs SOO in the right solution only send either On (e.g. upper FB) or Off (e.g. lower FB).

1.1.2.3 Toggle Mode

Toggle Mode denotes the behaviour where the value of the outputs SOO inverts on each transmission. The value of SOO can be calculated by the device internally or by interpreting the received value of the input InfoOnOff (IOO).

The concept of toggle mode is only meaningful in the realisation as given in Figure 3. In case the parameter Enable Toggle Mode (ETM) has the value Enable, the parameter On Off Action (OOA) becomes meaningless.

Toggle Mode implemented and active; IOO is not implemented

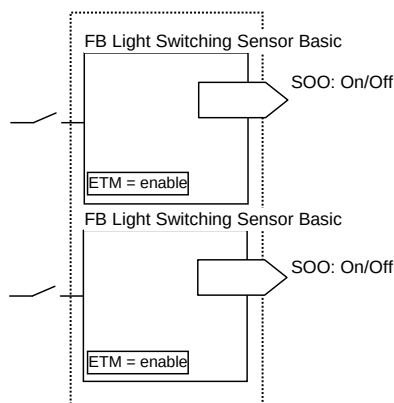


Figure 4 – Toggle Mode without IOO

Rule 1

Each subsequent transmission on the DP SOO will have an inverted value compared to the previous transmission.

$$SOO_{n+1} = \text{NOT}(SOO_n)$$

Toggle Mode implemented and active; IOO is implemented

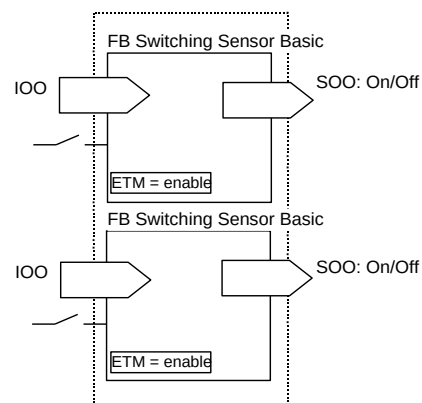


Figure 5 – Toggle Mode with IOO

Rule 2

Rule 1 is in this case **extended** with the following: on each reception of IOO, SOO becomes the value of IOO.

$$SOO = IOO$$

The relation between IOO and Toggle Mode (Figure 5) can be summarized in the following pseudo code.

```

if(Update(DP_IOO)) { // This is rule 2.
    DP_SOO = GetValue(DP_IOO);
}
if (UserInteraction) { // This is rule 1.
    DP_SOO = InvertValue(DP_SOO);
    SendValue(DP_SOO);
}

```

1.1.2.4 Functionality of the HMI

No requirements are specified concerning the interpretation of the HMI.

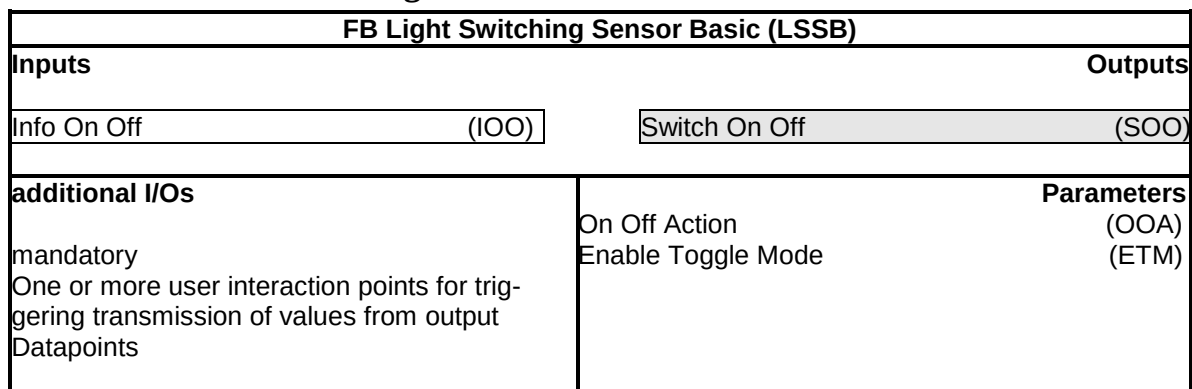
The following Datapoints shall allow coping with various HMI flavours and approaches towards controlling switching actuators.

1. Input IOO
2. Parameter Enable Toggle Mode
3. Parameter On Off Action

1.1.3 Constraints

This FB does not foresee any functionality for scene controlling. The inputs in the switching actuator for scene control can be controlled by a dedicated FB for scene purposes.

1.1.4 Functional Block Diagram



mandatory
 optional

Figure 6 – Functional Block Diagram for FB Light Switching Sensor Basic

1.1.5 Datapoints

Table 1 – Datapoint overview

Datapoint	Description/Remarks	Datapoint Type
Outputs		
Switch On Off	To switch the lamp on (=1) or off (=0)	DPT_Switch (1.001)
Inputs		
Info On Off	To indicate the status of the lamp	DPT_Switch (1.001)
Parameters		
On Off Action	Limits the values sent by the output SOO to only On or Off	DPT_Switch (1.001)
Enable Toggle Mode	Specifies whether the output SOO is the inverse of the input IOO or not.	DPT_Enable (1.003)

1.1.6 FB Profiles ¹⁾

Features and options	Basic FB	Standard Mode
		LSSB Profile 1
Output SOO	M	GO
Input IOO	O	(GO)

Figure 7: Runtime Interworking – Dependence on Configuration Mode

Table 2 - Parameters

		Basic FB	S-Mode
Parameters	OOA	O	O
	ETM	O	O

As regards the implementation of Parameters in Standard Mode or the LTE-Mode Standard Mode Interface, in case of memory mapped Datapoints the DPT may be implementation specific. In case of implementation as a Property of an Interface Object, the use of the standardised parameter specification is mandatory.

¹⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

1.1.7 Detailed specification of Datapoints

1.1.7.1 Output Switch On Off

DP Name:	Switch On Off	Abbr.:	SOO	Mandatory	☒
FB Name:	Light Switching Sensor Basic (LSSB)	Can be internal	☐		
Description					
The output Switch On Off shall be used to control the switching status of the switching actuator.					
Datapoint Type					
DPT Name:	DPT_Switch				
DPT Format:	B ₁	DPT ID:	1.001		
Field	Description	Supp.	Range	Unit	Default
b	This field shall indicate whether the switching actuator will switch on (1) or off (0)	M	{0, 1}	-	-
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐		☐
Transmit on bus (only for output):		☐	Read from bus (only for input):		☐
Exception Handling					
Through the parameter <u>On Off Action</u> it is possible that on human interaction only one value of the range is transmitted					
Special Features					
None.					

1.1.7.2 Input Info On Off

DP Name:	Info On Off		IOO	Mandatory	☐
FB Name:	Light Switching Sensor Basic (LSSB)			Can be internal	☐
Description					
The input „Info On Off“ shall be used to receive the status of the switching actuator. This information can be used solely for visualisation purposes, for realising the toggle functionality or other purposes.					
Datapoint Type					
DPT_Name:	DPT_Switch				
DPT_Format:	B ₁	DPT_ID:	1.001		
Field	Description	Supp.	Range	Unit	Default
b	Indicates the status of the switching actuator.	M	{0, 1}	-	-
Access Type					
◆ Input					
	N → this	☐	1 → this	☒	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	---			
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Actual value:
		Transmit on bus (only for output):		☐	Read from bus (only for input):
	☐				
Exception Handling					
If this DP is not received (communication failure or configuration mistake) and the toggle functionality is implemented, then the specification the output SOO will still toggle, as specified in 1.1.2.3.					
Special Features					
None.					

1.1.7.3 Parameter On Off Action

FB:	Light Switching Sensor Basic (LSSB)	Property Name (Server):	On Off Action (OOA)	Mandatory	☐
				Optional	☒
Description:					
If this parameter is implemented, then SOO shall always send one single value (Off or On). This parameter only makes sense in certain realisation flavours of this FB: see 1.1.2.2.					
DPT:	Name	DPT_Switch	DPT ID	1.001	Datatype format
					B ₁
Field	Description	Sup.	Range	Unit	Default
b	0 = Off SOO shall only send value '0'	M	{0,1}	None	None
	1 = On SOO shall only send value '1'				
Communication:					
DP Address:	object_type:	421	PID:	51	
(in the server)	start_index:	1	nr_of_elem:	1	
Property access:	Read only	☐	Read/Write	☒	
Protection	Read level	-	Write level	-	
Exception Handling:	Value after Power-up:	Stored Value	☒	Act Value	☐
				Default Value	☐
None.					
Special Features:					
None.					

1.1.7.4 Parameter Enable Toggle Mode

FB:	Light Switching Sensor Basic (LSSB)	Property Name (Server):	Enable Toggle Mode (ETM)	Mandatory	☐		
Optional ☒							
Description:							
If this parameter has the value “Enable” then the value of the output SOO shall toggle each time it is transmitted; if this parameter has the value “Disable” then the output SOO shall not be toggled. See 1.2.2.3.							
DPT:	Name	DPT_Enable	DPT ID	1.003	Datatype format	B ₁	
Field	Description			Sup.	Range	Unit	Default
b	Enables or disables Toggle Mode.			M	{0,1}	None	None
Communication:							
DP Address:		object_type:	421	PID:		52	
(in the server)		start_index:	1	nr_of_elem:		1	
Property access:		Read only ☐	Read/Write ☒				
Protection		Read level	-	Write level		-	
Exception Handling:		Value after Power-up:	Stored Value ☒	Act Value ☐		Default Value ☐	
None.							
Special Features:							
None.							

1.2 FB Dimming Sensor Basic (Object Type 420)

1.2.1 Aims and objectives

The FB Dimming Sensor Basic is used in the Application Domain Lighting for providing input data to dimming actuators. It specifies the functionality, for example contained in a switch or a push button, to increase or decrease the brightness of a lamp or to switch the lamp on/off.

Display elements (with own FBs) can be integrated to show the status of the dimming actuator.

The inputs and outputs of the Functional Block are described but not the Human Machine Interface (HMI). Consequently, the manufacturers of the button or switch have the opportunity to implement their design and their operation methods.

1.2.2 Functional specification

1.2.2.1 Overview

The FB Dimming Sensor Basic provides hardwired inputs for triggering transmission of values of output Datapoints.

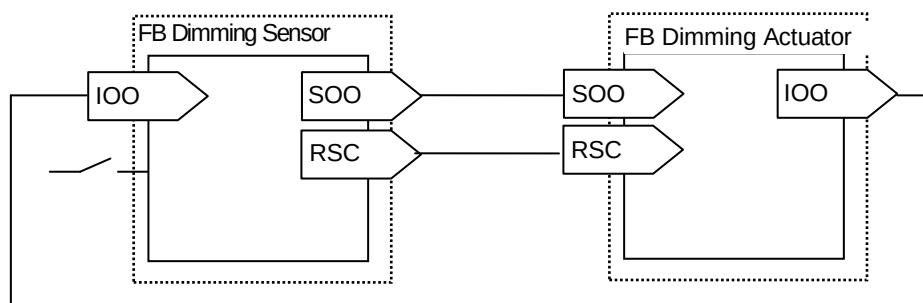


Figure 8 – Basic communication model (example with Relative Dimming Sensor)

Figure 8 gives an application example in which the lighting is controlled relatively: the set value of the dimmer is increased and decreased in relative steps starting from the current value. It is also possible to control the set value of the FB Dimming Actuator directly through the DP Absolute Setvalue Control (ASC), as shown in Figure 9. Both models can be combined in a sensor implementation.

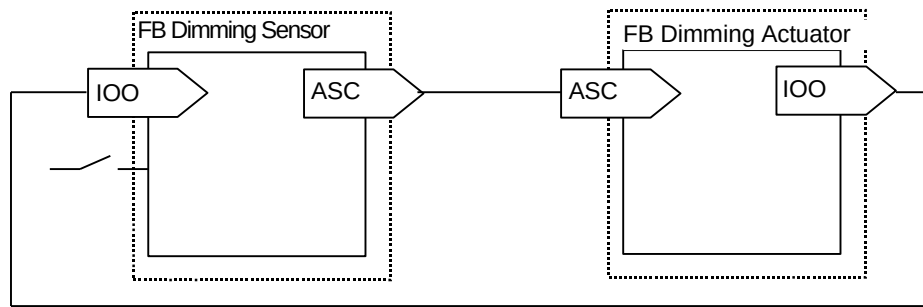


Figure 9 – Basic communication model (example with Absolute Dimming Sensor)

1.2.2.2 Parameter On Off Action

The parameter On Off Action shall limit the values transmitted by the outputs SOO and RSC to either only 0 (“Off”) and “Dimming Down/Stop” or 1 (“On”) and “Dimming Up/Stop”.

This mainly makes sense if an appliance is realised as a combination of two FBs DSB each with one interaction point. This realisation however is only meaningful if the parameter ETM is not implemented or is set to the value “Disable”.

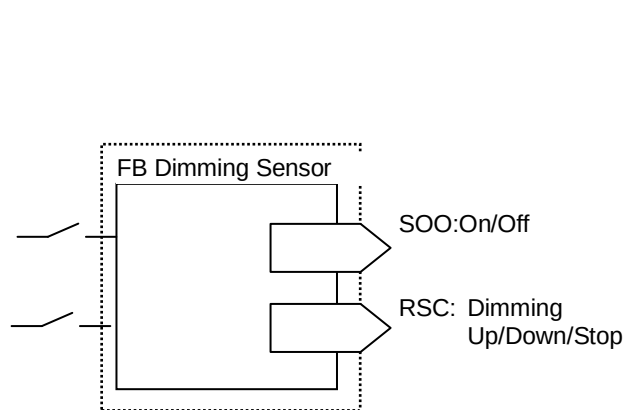


Figure 10 – One FB with two interaction points

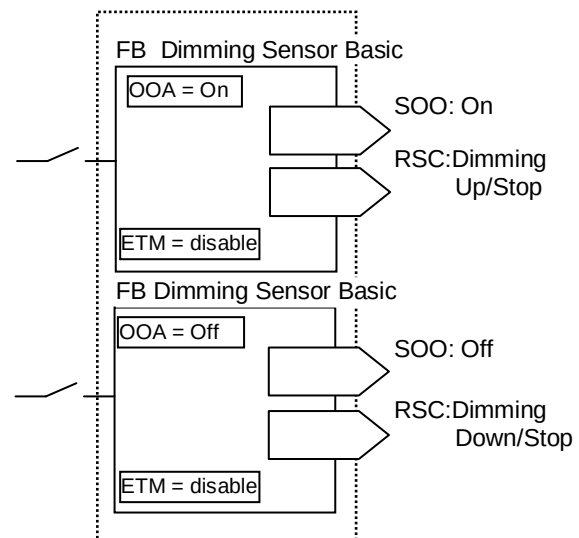


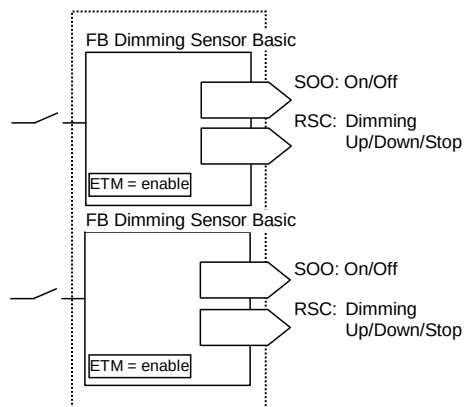
Figure 11 – Two FBs each with one interaction points

The dashed outlined areas realise the same functionality. The DP SOO in the left solution sends both On and Off. The DPs SOO in the right solution only send either On (e.g. upper FB) or Off (e.g. lower FB).

1.2.2.3 Toggle Mode

Toggle Mode denotes the behaviour where the value of the output SOO inverts on each transmission. The value of SOO can be calculated by the device internally or by interpreting the received value of the input InfoOnOff (IOO).

The concept of toggle mode is only meaningful in the realisation as given in Figure 11. In case the parameter Enable Toggle Mode (ETM) is set to the value Enable, the parameter On Off Action (OOA) becomes meaningless.

**Toggle Mode implemented and active;
IOO is not implemented****Figure 12 – Toggle Mode without IOO****Rule 1**

Each subsequent transmission on the DP SOO will have an inverted value compared to the previous transmission.

$$SOO_{n+1} = \text{NOT}(SOO_n)$$

The relation between IOO and Toggle Mode (Figure 13) can be summarized in the following pseudo code.

```

if(Update(DP_IOO)) { // This is rule 2.
    DP_SOI = GetValue(DP_IOO);
}
if (UserInteraction) { // This is rule 1.
    DP_SOI = InvertValue(DP_SOI);
    SendValue(DP_SOI);
}

```

The interpretation of Toggle Mode concerning DP RSC and DP ASC is manufacturer specific.

1.2.2.4 Functionality of the HMI

No requirements are specified concerning the interpretation of the HMI.

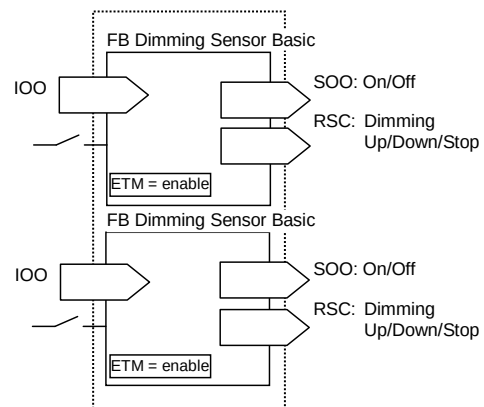
The following parameters shall allow coping with various HMI flavours and approaches towards controlling dimming actuators.

1. Input IOO
2. Parameter Enable Toggle Mode
3. Parameter On Off Action

1.2.3 Constraints

This FB does not foresee any functionality for scene controlling. The inputs in the dimming actuator for scene control can be controlled by a dedicated FB for scene purposes.

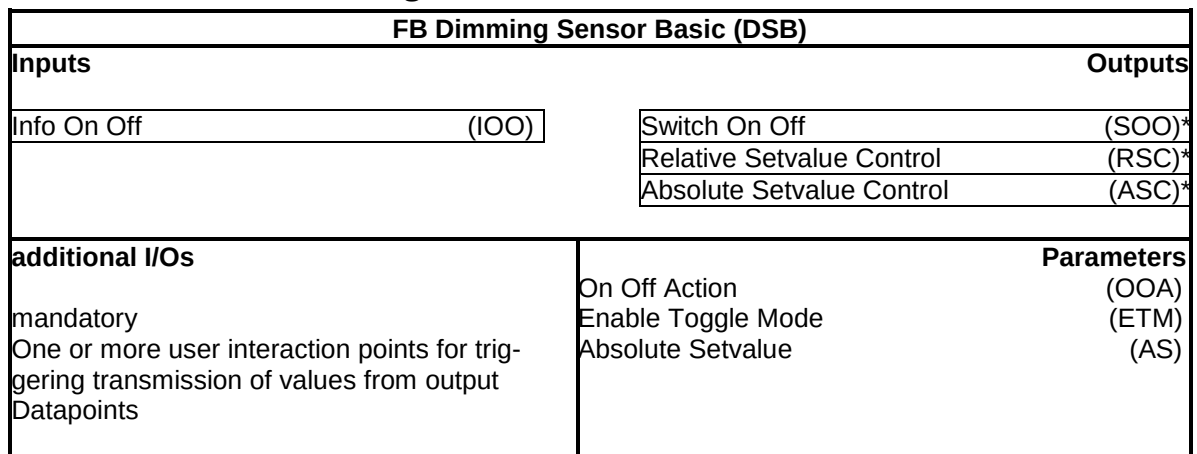
The step size for the DP Relative Setvalue Control is manufacturer specific. No standard parameter is foreseen to this purpose.

**Toggle Mode implemented and active;
IOO is implemented****Figure 13 – Toggle Mode with IOO****Rule 2**

Rule 1 is in this case **extended** with the following: on each reception of IOO, SOO becomes the value of IOO.

$$SOO = IOO$$

1.2.4 Functional Block Diagram



 mandatory  optional

* For the allowed combination of output Datapoints see Table 4.

Figure 14 – Functional Block Diagram for FB Dimming Sensor Basic

1.2.5 Datapoints

Table 3 – Datapoint overview

Datapoint	Description/Remarks	Datapoint Type
Outputs		
Switch On Off	To switch the lamp on (=1) or off (=0)	DPT_Switch (1.001)
Relative Setvalue Control	To dim the brightness of the lamp up or down.	DPT_Control_Dimming (3.007)
Absolute Setvalue Control	To directly set the Setvalue of the dimming actuator.	DPT_Scaling (5.001)
Inputs		
Info On Off	To indicate the status of the lamp	DPT_Switch (1.001)
Parameters		
On Off Action	Limits the values sent by the output SOO / RSC to only On/Dimming Up or Off/Dimming Down	DPT_Switch (1.001)
Enable Toggle Mode	Specifies whether the output SOO is the inverse of the input IOO or not.	DPT_Enable (1.003)
Absolute Setvalue	Value for the output Absolute Setvalue Control	DPT_Scaling (5.001)

1.2.6 FB Profiles ²⁾

Features and options	Basic FB	Standard Mode		LTE-Mode
		FB Profile 1	FB Profile 2	Standard Mode Interface
Input IOO	O	(GO)	(GO)	(GO)
Output SOO	O	GO	(GO)	GO
Output RSC	O	GO	(GO)	(GO)
Output ASC	O	(GO)	GO	(GO)

DSB FB Profile 1 models the Relative Dimming Sensor; *DSB FB Profile 2* models the Absolute Dimming Sensor. Table 4 exclusively specifies the allowed combinations of output Datapoints. Combination 1 is for the Relative Dimming Sensor, combination 2 is for the Absolute Dimming Sensor.

Table 4 - Combination of output Datapoints in a Dimming Sensor

Datapoints	Combination	
	Relative Dimming Sensor	Absolute Dimming Sensor
	1	2
SOO	M	O
RSC	M	O
ASC	O	M

Table 5 - Parameters

		Basic FB	S-Mode
Parameters	OOA	O	O
	ETM	O	O
	AS	O	O

As regards the implementation of Parameters in Standard Mode or the LTE-Mode Standard Mode Interface, in case of memory mapped Datapoints the DPT may be implementation specific. In case of implementation as a Property of an Interface Object, the use of the standardised parameter specification is mandatory.

²⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

1.2.7 Detailed specification of Datapoints

1.2.7.1 Output Switch On Off

DP Name:	Switch On Off	Abbr.:	SOO	Mandatory	☐ a)
FB Name:	Dimming Sensor Basic (DSB)	Can be internal			☐
Description					
The output Switch On Off shall be used to control the switching status of the dimming actuator.					
Datapoint Type					
DPT Name:	DPT_Switch				
DPT Format:	B ₁	DPT ID:	1.001		
Field	Description	Supp.	Range	Unit	Default
b	This field shall indicate whether the dimming actuator will switch on (1) or off (0)	M	{0, 1}	-	-
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐		☐
Transmit on bus (only for output):		☐	Read from bus (only for input):		☐
Exception Handling					
Through the parameter <u>On Off Action</u> it is possible that on human interaction only one value of the range is transmitted					
Special Features					
None.					

a) For allowed combinations of output Datapoints see Table 4

1.2.7.2 Output Relative Setvalue Control

DP Name:	Relative Setvalue Control	Abbr.:	RSC	Mandatory	☐ ^{a)}
FB Name:	Dimming Sensor Basic (DSB)			Can be internal	☐
Description					
This Output shall be used to increase or decrease the brightness (StepCode ≠ 0) of the dimming actuator or to stop the ongoing dimming action (StepCode = 0). According to the requirements concerning the support of ranges by Output DPTs as specified in Chapter 3/7/1 "Interworking Model" clauses 2.1.7.2.1.1 and 2.4 it is allowed that the range of the StepCode be limited to values smaller than 111b. The value 000b shall however always be supported.					
Datapoint Type					
DPT_Name:	DPT_Control_Dimming				
DPT_Format:	B ₁ U ₃	DPT_ID:	3.007		
Field	Description	Supp.	Range	Unit	Default
c	Shall specify whether the dimming set value shall be increased or decreased.	M	{0,1}	none	-
StepCode	Shall specify the dimming step size.	M	[000b ... 111b]	none	-
Access Type					
♦ Output					
☐ this → M ☐ Spontaneous ☐ Request	☒ this → 1 ☐ COV: ☐ Cyclic	☐ Δ-Value: ☐ Period:	☐ Min repetition time:		
Communication Type					
♦ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Through the parameter <u>On Off Action</u> it is possible that on human interaction only one value of the range of the field c is transmitted. This way, it can be configured that the sensor only dims up/stops the dimming value or dims down/stops the dimming value.					
Special Features					

^{a)} For allowed combinations of output Datapoints see Table 4

1.2.7.3 Output Absolute Setvalue Control (ASC)

DP Name:	Absolute Setvalue Control		Abbr.:	ASC	Mandatory	☐ ^{a)}
FB Name:	Dimming Sensor Basic (DSB)				Can be internal	☐
Description						
This output shall be used to directly set the setvalue of the dimming actuator. The algorithm to calculate the value of this output is not standardised. It may be given through the parameter <u>Absolute Setvalue</u>. If the parameter On Off Action is implemented and has the value "Off", then it is <i>recommended</i> that this output value only sends 0 %.						
Datapoint Type						
DPT_Name:	DPT_Scaling					
DPT_Format:	U ₈		DPT_ID:	5.001		
Field	Description	Supp.	Range	Unit	Default	
UnsignedValue	The set value for the dimming actuator.	M	0 % to 100 %	%	-	
Access Type						
◆ Output						
this → M	☒	this → 1	☐			
Spontaneous	☒	COV:	☐	Δ-Value:		
		Cyclic	☐	Period:		
Request	☐					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:	---					
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
None.						
Special Features						
None.						

^{a)} For allowed combinations of output Datapoints see Table 4

1.2.7.4 Input Info On Off

DP Name:	Info On Off		IOO	Mandatory	☐
FB Name:	Dimming Sensor Basic (LSSB)			Can be internal	☐
Description					
The input „Info On Off“ shall be used to receive the switching status of the dimming actuator. This information can be used solely for visualisation purposes, for realising the toggle functionality or other purposes.					
Datapoint Type					
DPT_Name:	DPT_Switch				
DPT_Format:	B ₁	DPT_ID:	1.001		
Field	Description	Supp.	Range	Unit	Default
b	Indicates the status of the dimming actuator 0 = dimming actuator off 1 = dimming actuator on	M	{0, 1}	-	-
Access Type					
◆ Input					
	N → this	☐	1 → this	☒	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	---			
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
		Saved value:	☐	Actual value:	
	Transmit on bus (only for output):		☐	Read from bus (only for input): ☐	
Exception Handling					
If this DP is not received (communication failure or configuration mistake) and the toggle functionality is implemented, then the specification the output SOO will still toggle, as specified in 1.2.2.3.					
Special Features					
None.					

1.2.7.5 Parameter On Off Action

FB:	Dimming Sensor Basic (LSSB)	Property Name (Server):	On Off Action (OOA)	Mandatory ☐	Optional ☒
Description:					
If this parameter is implemented, then SOO and RSC shall always send one single value (Off respectively Dimming Down/Stop or On respectively Dimming Up/Stop). This parameter only makes sense in certain realisation flavours of this FB: see 1.2.2.2. The influence of this Parameter on the Output ASC is not standardised.					
DPT:	Name	DPT_Switch	DPT ID	1.001	Datatype format B ₁
Field	Description			Sup.	Range Unit Default
b	0 = Off SOO shall only send value '0' RSC shall only send value 'dimming down/stop' 1 = On SOO shall only send value '1' RSC shall only send value 'dimming up/stop'			M	{0,1} None None
Communication:					
DP Address:		object_type:	420	PID:	51
(in the server)		start_index:	1	nr_of_elem:	1
Property access:		Read only ☐	Read/Write ☒		
Protection		Read level	-	Write level	-
Exception Handling:		Value after Power-up:	Stored Value ☒	Act Value ☐	Default Value ☐
None.					
Special Features:					
None.					

1.2.7.6 Parameter Enable Toggle Mode

FB:	Dimming Sensor Basic (LSSB)	Property Name (Server):	Enable Toggle Mode (ETM)	Mandatory ☐	Optional ☒
Description:					
If this parameter has the value "Enable" then the value of the output SOO shall toggle each time it is transmitted; if this parameter has the value "Disable" then the output SOO shall not be toggled. See 1.2.2.3.					
DPT:	Name	DPT_Enable	DPT ID	1.003	Datatype format B ₁
Field	Description			Sup.	Range Unit Default
b	Enables or disables Toggle Mode.			M	{0,1} None None
Communication:					
DP Address:		object_type:	420	PID:	52
(in the server)		start_index:	1	nr_of_elem:	1
Property access:		Read only ☐	Read/Write ☒		
Protection		Read level	-	Write level	-
Exception Handling:		Value after Power-up:	Stored Value ☒	Act Value ☐	Default Value ☐
None.					
Special Features:					
None.					

1.2.7.7 Parameter Absolute Setvalue

FB:	Dimming Sensor Basic (DSB)	Property Name (Server):	Absolute Setvalue (AS)	Mandatory	☐	Optional	☒
Description:							
This parameter shall define the value that shall be sent with the output Absolute Setvalue Control (ASC).							
DPT:	Name	DPT_Scaling	DPT ID	5.001	Datatype format	U ₈	
Field	Description			Sup.	Range	Unit	Default
UnsignedValue	Contains the value for ASC.			M	cs	%	-
Communication:							
DP Address:	object_type:	420	PID:	53			
(in the server)	start_index:	1	nr_of_elem:	1			
Property access:	Read only	☐	Read/Write	☒			
Protection	Read level	-	Write level	-			
Exception Handling:	Value after Power-up:	Stored Value	☒	Act Value	☐	Default Value	
☐							
None.							
Special Features:							
None.							

1.3 FB Colour Temperature Setting Sensor (Object Type 426)

1.3.1 Aims and objectives

The control of the colour of white light, named ‘tunable white’, considers the possibility to generate light quality for different human expectations respective to “biological light” for better good health.

1.3.2 Introduction and overview

The FB Colour Temperature Setting Sensor is used in the Application Domain Lighting for providing input data to colour temperature actuators. It specifies the functionality, for example contained in a switch or a push button, to increase or decrease the colour temperature of a lamp.

The implementation of this functionality is not restricted to push buttons only. It may for instance be implemented in a larger controller, scheduler, gateway...

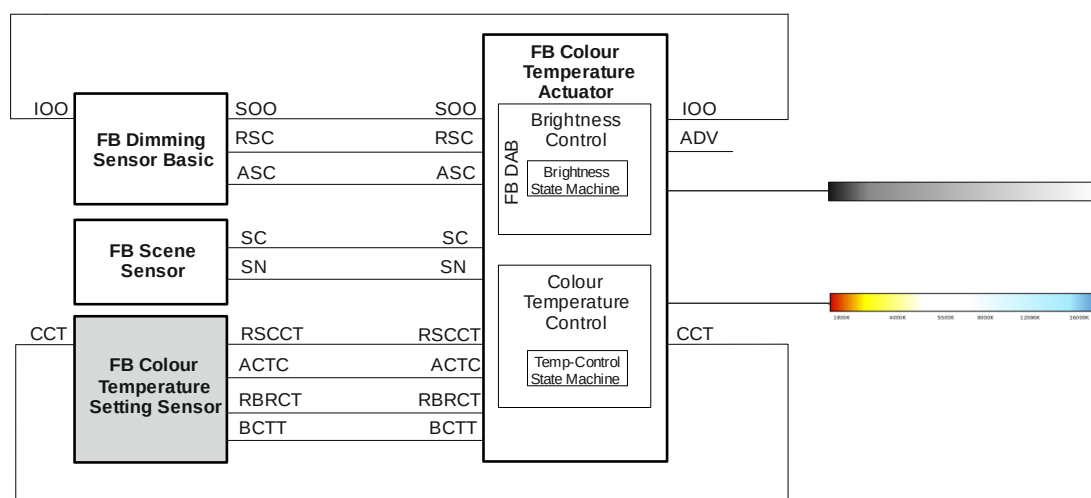


Figure 15 – Overview of the Application Model Colour Temperature Control in the Application Domain Lighting

The colour temperature is defined as absolute temperature (in Kelvin) of a black body. This black body changes its colour from “red” through “yellow” to “white” as its temperature increases (black body line – BBL).

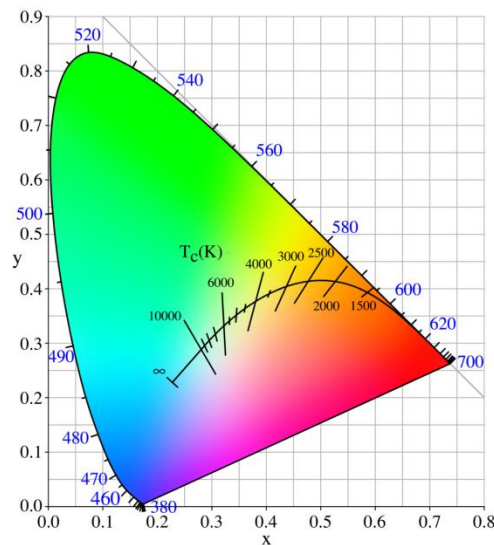


Figure 16 - Control of colour temperature along the BBL (black body line)

The total output light intensity depends on the requested colour temperature and the brightness.

Many light sources do not emit a light colour caused by the temperature of the light source itself and do not follow exactly the black body line (BBL).

EXAMPLE 1 LED or fluorescent lamp versus tungsten lamp.

The colour emitted from the light source that is perceived by the human eye and that matches closest to black body line is called Correlated Colour Temperature (CCT).

In the context of Colour Temperature “TC” refers to the colour closest to the BBL aligned to tint line (Figure 16).

The colour temperature range covers the area from “cold” blue colour (20 000 K) to warm red colour (1 000 K).



Figure 17 - Colour Temperature range

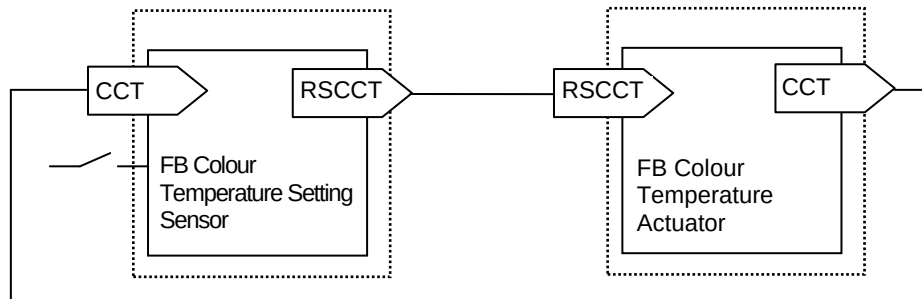
Display elements (with own FBs) can be integrated to show the status of the colour temperature actuator.

The Inputs and Outputs of the Functional Block are described, but not the Human Machine Interface (HMI). Consequently, the manufacturers of the button or switch have the opportunity to implement their design and their operation methods.

1.3.3 Functional specification

1.3.3.1 Overview

The FB Colour Temperature Setting Sensor may provide hardwired inputs for triggering transmission of values of output Datapoints.



**Figure 18 – Basic communication model
(example with Relative Colour Temperature Setting Sensor)**

Figure 18 gives an application example in which the colour temperature is controlled relatively: the set value of the colour temperature is increased and decreased in relative steps to the full range.

Additionally the optional DP *Relative Brightness Relative Colour Temperature Control* (DP RBRCT) shall allow controlling brightness and colour temperature in a simultaneous way.

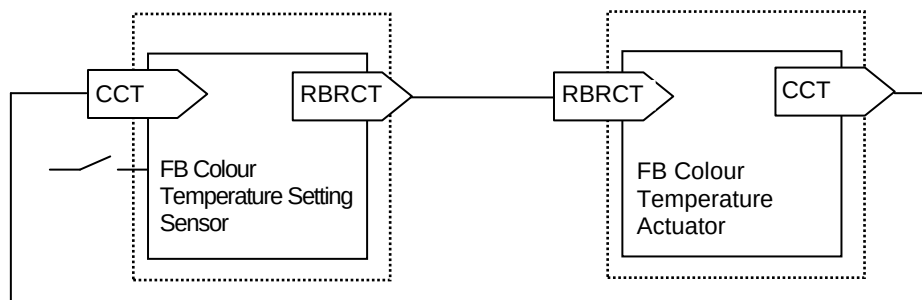


Figure 19 –Communication model (example with combined relative control of brightness and colour temperature)

The DP *Absolute Colour Temperature Control* (DP ACTC) shall serve to control the set value of the FB Colour Temperature Actuator directly, as shown in Figure 20.

Both models (relative - and absolute temperature control) can be combined in a sensor implementation.

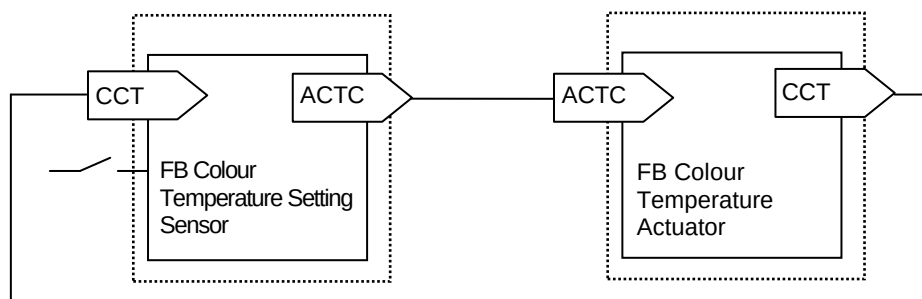


Figure 20 – Communication model (example with Absolute Colour Temperature Setting Sensor)

It shall also be possible to control the transition of the colour temperature for dynamic aspects.

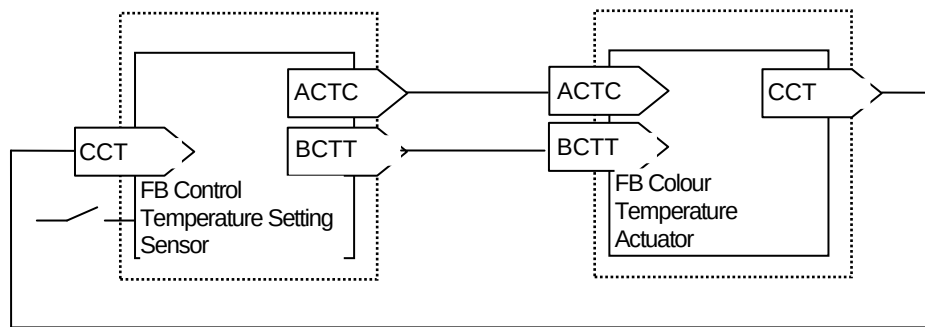


Figure 21 – Communication model: Colour Control Sensor with Colour Control Actuator

Figure 21 gives an example in which the colour is controlled from the Colour Temperature Setting Sensor (DP ACTC). In addition it should be possible to control the colour temperature transition (DP ACT_TT).

1.3.3.2 Functionality of the HMI

No requirements are specified concerning the interpretation of the HMI.

1.3.4 Constraints

- This FB does not foresee any functionality for scene controlling. The inputs in the colour temperature actuator for scene control can be controlled by a dedicated FB for scene purposes.
- The step sizes for the DP Relative Setvalue Control and for DP Relative Brightness / Colour Temperature Control is manufacturer specific. No standard parameter is foreseen to this purpose.

1.3.5 Functional Block Diagram

FB Colour Temperature Setting Sensor (CTSS)		426
Inputs	Outputs	
Current Colour Temperature (CCT)	Relative Setvalue Control Colour Temperature (RSCCT)*	
	Absolute Colour Temperature Control (ACTC)*	
	Relative Brightness Relative Colour Temperature Control (RBRCT)	
	Brightness Colour Temperature Transition (BCTT)	
additional I/Os	Parameters	
Mandatory	Absolute Colour Temperature Control Value (ACT_CV)	
One or more user interaction points for triggering transmission of values from output Datapoints.	Absolute Colour Temperature Transition Time (ACT_TT)	

mandatory optional

* For the allowed combination of output Datapoints see 1.3.7.

Figure 22 – Functional Block Diagram for FB Colour Temperature Setting Sensor

1.3.6 Datapoints

Table 6 – Datapoint overview

Datapoint	Description/Remarks	Datapoint Type
Outputs		
Relative Setvalue Control Colour Temperature	To relative control the colour temperature of the lamp up or down.	DPT_Control_Dimming (3.007)
Absolute Colour Temperature Control	To directly set (absolute control) the colour temperature of the lamp.	DPT_Absolute_Colour_Temperature (7.600)
Relative Brightness Relative Colour Temperature	To control simultaneous the brightness and colour temperature of the lamp.	DPT_Brightness_Colour_Temperature_Control (250.600)
Brightness Colour Temperature Transition	To cause a transition of the brightness and the colour temperature of the lamp within a time period.	DPT_Brightness_Colour_Temperature_Transition (249.600)
Inputs		
Current Colour Temperature	To receive the current colour temperature from the actuator	DPT_Absolute_Colour_Temperature (7.600)
Parameters		
Absolute Colour Temperature Control Value	Value for the Output Absolute Colour Temperature Control	DPT_Absolute_Colour_Temperature (7.600)
Absolute Colour Temperature Transition Time	Transition Time Colour Temperature transition	DPT_TimePeriod100MS ec (7.004)

1.3.7 FB Profiles ³⁾

		Standard Mode		LTE-Mode
		FB Profile 1	FB Profile 2	Standard Mode Interface
Features and options	Basic FB			
Input CCT	O	(GO)	(GO)	(GO)
Output RSCCT	O	GO	(GO)	(GO) ^a
Output ACTC	O	(GO)	GO	(GO) ^a
Output RBRCT	O	(GO)	(GO)	(GO)
Output BCTT	O	(GO)	(GO)	(GO)
^a At least one of these DP's shall be implemented.				

CTSS FB Profile 1 models the Relative Colour Temperature Setting Sensor; CTSS FB Profile 2 models the Absolute Colour Temperature Setting Sensor. Table 7 exclusively specifies the allowed combinations of output Datapoints. Combination 1 is for the Relative Colour Temperature Setting Sensor, combination 2 is for the Absolute Colour Temperature Setting Sensor.

³⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

Table 7 - Combination of output Datapoints in a Colour Temperature Setting Sensor

Datapoints	Combination	
	Relative Colour Temperature Setting Sensor	Absolute Colour Temperature Setting Sensor
	1	2
RSCCT	M	O
ACTC	O	M
RBRCT	O	O
BCTT	O	O

As regards the implementation of Parameters in Standard Mode or the LTE-Mode Standard Mode Interface, in case of memory mapped Datapoints the DPT may be implementation specific. In case of implementation as a Property of an Interface Object, the use of the standardised parameter specification is mandatory.

1.3.8 Detailed specification of Datapoints

1.3.8.1 Output Relative Setvalue Control Colour Temperature (RSCCT)

DP Name:	Relative Setvalue Control Colour Temperature		Abbr.:	RSCCT	Mandatory	☐ a)																										
FB Name:	Colour Temperature Setting Sensor (CTSS)				Can be internal	☐																										
Description																																
This Output shall be used to increase or decrease the colour temperature (StepCode ≠ 0) of the colour temperature actuator or to stop the ongoing colour temperature change (StepCode = 0). According the requirements concerning the support of ranges by Output DPTs as specified in Chapter 3/7/1 "Interworking Model" clauses 2.1.7.2.1.1 and 2.4 it is allowed that the range of the StepCode be limited to values smaller than 111b. The value 000b shall however always be supported.																																
Datapoint Type																																
DPT_Name:	DPT_Control_Dimming																															
DPT Format:	B ₁ U ₃		DPT_ID:	3.007																												
Field	Description	Supp.	Range	Unit	Default																											
c	Shall specify whether the colour temperature value shall be increased or decreased.	M	{0,1}	none	-																											
StepCode	Shall specify the fading step size or the code to stop the fading	M	000b to 111b	none	-																											
Access Type																																
◆ Output																																
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☐</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☐			Δ-Value:				Cyclic	☐			Period:		Request	☐			<table border="1"> <tr> <td>Min repetition time:</td> <td></td> </tr> </table>						Min repetition time:	
this → M	☒	this → 1	☐																													
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Power down:	Save:	☐																														
Power up:	Value:	No initialisation:	☒	Default value:	☐																											
		Saved value:	☐	Current value (not for input):	☐																											
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																											
Exception Handling																																
None.																																
Special Features																																
None.																																

a) For allowed combinations of output Datapoints see Table 7.

1.3.8.2 Output Absolute Colour Temperature Control (ACTC)

DP Name:	Absolute Colour Temperature Control		Abbr.:	ACTC	Mandatory	☐ ^{a)}
FB Name:	Colour Temperature Setting Sensor (CTSS)				Can be internal	☐
Description						
This Output shall be used to directly set the colour temperature value of a lamp. The algorithm to calculate the value of this output is not standardised. It may be given through the parameter Absolute Colour Temperature Control Value.						
Datapoint Type						
DPT_Name:	DPT_Absolute_Colour_Temperature					
DPT_Format:	U ₁₆		DPT_ID:	7.600		
Field	Description	Supp.	Range	Unit	Default	
Absolute Colour Temperature	The colour temperature for the lamp.	M	0 K to 65 535	K	none	
Access Type						
♦ Output						
this → M	☒	this → 1	☐			
Spontaneous	☒	COV:	☐	Δ-Value:		
		Cyclic	☐	Period:		
Request	☐					
Communication Type						
♦ Group Object Datapoint					Mandatory:	☒
Default Group Address:	---					
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐		
Exception Handling						
None.						
Special Features						
None.						

^{a)} For allowed combinations of output Datapoints see Table 7.

1.3.8.3 Output Relative Brightness Relative Colour Temperature (RBRCT)

DP Name:	Relative Brightness Relative Colour Temperature	Abbr.:	RBRCT	Mandatory	☐ a)
FB Name:	Colour Temperature Setting Sensor (CTSS)			Can be internal	☐
Description					
This Output shall be used to simultaneously control the brightness and the colour temperature (StepCodes ≠ 0) of the lamp or to stop the ongoing change (StepCodes = 0) of the brightness or the colour temperature. According the requirements concerning the support of ranges by Output DPTs as specified in Chapter 3/7/1 "Interworking Model" clauses 2.1.7.2.1.1 and 2.4 it is allowed that the ranges of the StepCodes be limited to values smaller than 111b. The values 000b shall however always be supported.					
Datapoint Type					
DPT Name:	DPT_Brightness_Colour_Temperature_Control				
DPT Format:	r4B1U3r4B1U3B8	DPT ID:	250.600		
Field	Description	Supp.	Range	Unit	Default
CCT	Shall specify whether the colour temperature value shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Colour Temperature	Shall specify the step size or the code to stop the change.	M	implementation specific. 000b shall be supported.	None.	None.
CB	Shall specify whether the brightness value shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Brightness	The amount of intervals in which the range of 0 % to 100 % is subdivided, or the break indication.	M	implementation specific. 000b shall be supported.	None.	None.
Masks: b1	This field shall indicate whether the fields CCT and Step Code Colour Temperature are valid or not.	M	{0,1}	None.	None.
Masks: b0	This field shall indicate whether the fields CB and Step Code Brightness are valid or not.	M	{0,1}	None.	None.
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☐	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

a) For allowed combinations of output Datapoints see Table 7.

1.3.8.4 Output Brightness Colour Temperature Transition (BCTT)

DP Name:	Brightness Colour Temperature Transition	Abbr.:	BCTT	Mandatory	☐ ^{a)}																										
FB Name:	Colour Temperature Setting Sensor (CTSS)			Can be internal	☐																										
Description																															
This Output shall be used to control the transition for the brightness and for the colour temperature of the lamp. The algorithm to calculate the values of this output is not standardised. It may be given through the parameter Absolute Colour Temperature Control Value. In the FB Colour Temperature Setting Sensor, there is no standard Parameter for the <i>Absolute Brightness</i> in this DP: this shall be inherited from the Parameter <i>Absolute Setvalue</i> of the <i>FB Dimming Sensor Basic</i>.																															
Datapoint Type																															
DPT_Name:	DPT_Brightness_Colour_Temperature_Transition																														
DPT Format:	U ₁₆ U ₁₆ U ₈ B ₈	DPT_ID:	249.600																												
Field	Description	Supp.	Range	Unit	Default																										
Time Period	Unsigned time value for calculating transition time.	M	0 s to 6553,5 s	100M sec	none																										
Absolute Colour Temperature	The colour temperature for the lamp.	M	0 K to 65 535 K	K	none																										
Absolute Brightness	The absolute brightness value for the lamp.	M	0 % to 100 %	%	none																										
Masks b2	b2: validity of the Time Period	M	{0,1}	none	none																										
Masks b1	b1: validity of the Absolute Colour Temperature	M	{0,1}	none	none																										
Masks b0	b0: validity of the absolute brightness	M	{0,1}	none	none																										
Access Type																															
◆ Output																															
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☐</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☐			Δ-Value:				Cyclic	☐			Period:		Request	☐			<table border="1"> <tr> <td>Min repetition time:</td> <td></td> </tr> </table>					Min repetition time:	
this → M	☒	this → 1	☐																												
Spontaneous	☒	COV:	☐																												
		Δ-Value:																													
		Cyclic	☐																												
		Period:																													
Request	☐																														
Min repetition time:																															
Communication Type																															
◆ Group Object Datapoint				Mandatory:	☒																										
Default Group Address:		---																													
Dynamics																															
Power down:	Save:	☐																													
Power up:	Value:	No initialisation:	☒	Default value:	☐																										
		Saved value:	☐	Current value (not for input):	☐																										
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																										
Exception Handling																															
None.																															
Special Features																															
None.																															

^{a)} For allowed combinations of output Datapoints see Table 7.

1.3.8.5 Input Current Colour Temperature (CCT)

DP Name:	Current Colour Temperature	CCT	Mandatory	☐
FB Name:	Colour Temperature Setting Sensor (CTSS)	Can be internal	☐	
Description				
The Input "Current Colour Temperature" shall be used to receive the Current Colour Temperature from the colour temperature actuator. This information can be used solely for visualisation or other purposes. NOTE 1 This Input does not give any indication whether the colour temperature actuator is switched on or off.				
Datapoint Type				
DPT_Name:	DPT_Absolute_Colour_Temperature			
DPT Format:	U ₁₆	DPT_ID:	7.600	
Field	Description	Supp.	Range	Unit Default
Absolute Colour Temperature	The Current Colour Temperature of the colour temperature actuator.	M	0 K to 65 535 K	K None.
Access Type				
◆ Input				
N → this	☐	1 → this	☒	
Spontaneous Request	☒	Cyclically:	☐	Time-out:
	☐	Polling:	☐	Period:
Communication Type				
◆ Group Object Datapoint				Mandatory: ☒
Default Group Address:	---			
Dynamics				
Power down:	Save:	☐		
Power up:	Value:	No initialisation:	☒	Default value:
		Saved value:	☐	Actual value:
	Transmit on bus (only for output):		☐	Read from bus (only for input):
Exception Handling				
None.				
Special Features				
None.				

1.3.8.6 Parameter Absolute Colour Temperature Control Value (ACT_CV)

FB:	Colour Temperature Setting Sensor (CTSS)	Property Name (Server):	Absolute Colour Temperature Control Value (ACT_CV)	Mandatory	☐
				Optional	☒
Description:					
This parameter shall define the value that shall be sent with the Outputs Absolute Colour Temperature Control (ACTC) and Brightness Colour Temperature Transition (BCTT).					
DPT:	Name	DPT_Absolute_Colour_Temperature	DPT ID	7.600	Datatype format U ₁₆
Field	Description	Sup.	Range	Unit	Default
Absolute Colour Temperature	Contains the value for ACTC and BCTT.	M	0 K to 65 535 K	K	None.
Communication:					
DP Address:	object_type:	426	PID:	100	
(in the server)	start_index:	1	nr_of_elem:	1	
Property access:	Read only	☐	Read/Write	☒	
Protection	Read level	-	Write level	-	
Exception Handling:	Value after Power-up:	Stored Value	☒	Act Value	☐
		Default Value	☐		
None.					
Special Features:					
None.					

1.3.8.7 Parameter Absolute Colour Temperature Transition Time (ACT_TT)

FB:	Colour Temperature Setting Sensor (CTSS)	Property Name (Server):	Absolute_ Colour Temperature Transition Time (ACT_TT)	Mandatory ☐	Optional ☒
Description:					
This parameter shall define the transition time that shall be sent with the output Current Colour Temperature Transition (BCTT).					
DPT:	Name	DPT_TimePeriod100Msec	DPT ID	7.004	Datatype format U ₁₆
Field	Description		Sup.	Range	Unit Default
TimePeriod	Unsigned time-value for calculating fading time.		M	100 ms to 65 535 ms	ms none
Communication:					
DP Address: (in the server)	object_type:	426	PID:	101	
	start_index:	1	nr_of elem:	1	
Property access:	Read only ☐	Read/Write ☒			
Protection	Read level	-	Write level	-	
Exception Handling:	Value after Power-up:	Stored Value ☒	Act Value ☐	Default Value ☐	
None.					
Special Features:					
None.					

1.4 FB Colour Setting Sensor RGB(W) (Object Type 424)

1.4.1 Aims and objectives

The control of the RGB Colour Setting considered the possibility to generate simple Colour Setting control, mostly for residential applications (e.g. RGB luminaire in a bathroom) or for other applications where non colour referenced luminaires are sufficient.

1.4.1.1 Introduction and overview

The FB Colour Setting Sensor RGB(W) is used in the Application Domain Lighting for providing input data to a colour actuator. It specifies the functionality, for example contained in a room control unit (e.g. with display for change of the colour with dedicated user interaction), to modify the colour of a lamp.

The implementation of this functionality is not restricted to push buttons only. It may for instance be implemented in a larger controller, scheduler, gateway ...

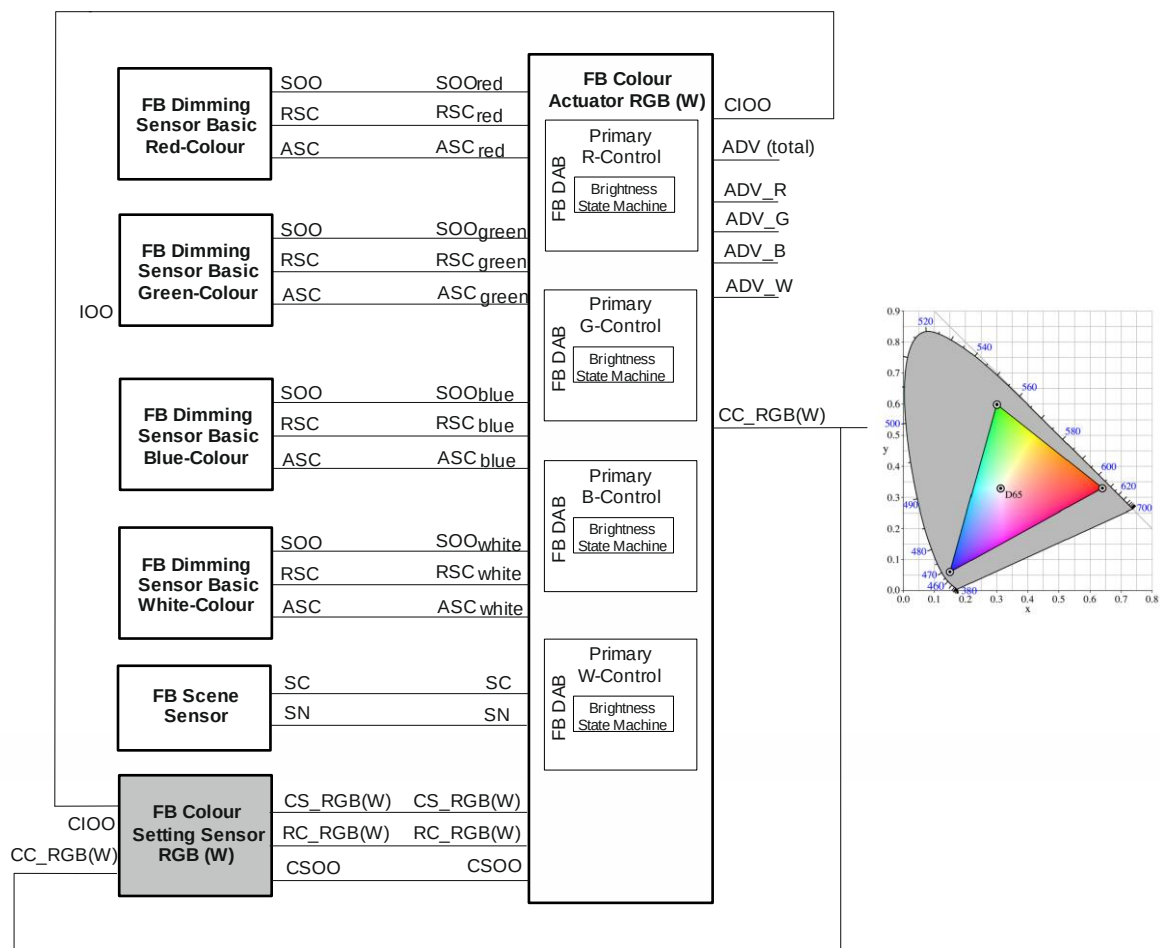


Figure 23 – Overview of the Application Model Colour RGB Control in the Application Domain Lighting

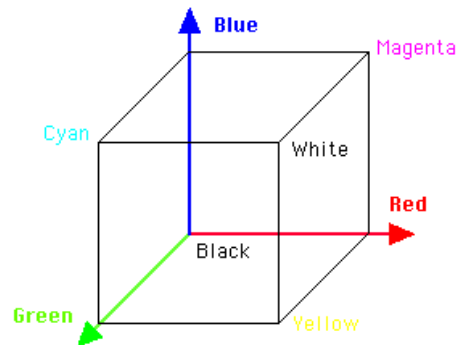
Respective to different colour coding models the FB shall at least support the two following models:

- RGB(W)

Other encodings and models may additionally be supported.

1.4.2 Definitions

Primary light output The colour of each primary light emitted by the lamp connected to the implementation of the FB CARGB.



Possible primaries are “red”, “green”, “blue” and “white”

light output brightness The brightness of the light emitted by the lamp connected to the implementation of the FB CARGB.

Primary set value The internal set value of the each primary light output

Combined light output This is the colour resulting from the combination of the Primaries.

This model has the following restrictions.

- The whole range of human colour perception is quite large. Not all possible colours can be represented. It may for instance require negative components to represent certain colours.
- Different colour reproducing devices have different “red”, “green”, “blue” primaries. The realised colour in the RGB colour model is device-dependent.

Summary for the use of the RGB colour model

- This is useful for simple colour control.
- Because of the device dependent interpretation of RGB, this coding is only suitable for point-to-point communication, this is, if there is only a single receiver.

1.4.2.1 Various

Display elements (with own FBs) can be integrated to show the status of the colour control actuator.

The Inputs and Outputs of the Functional Block are described, but not the Human Machine Interface (HMI). Consequently, the manufacturers of the colour control sensor devices have the opportunity to implement their design and their operation methods.

1.4.3 Functional specification

1.4.3.1 Overview

The FB Colour Setting Sensor RGB(W) provides hardwired inputs for triggering transmission of values of output Datapoints.

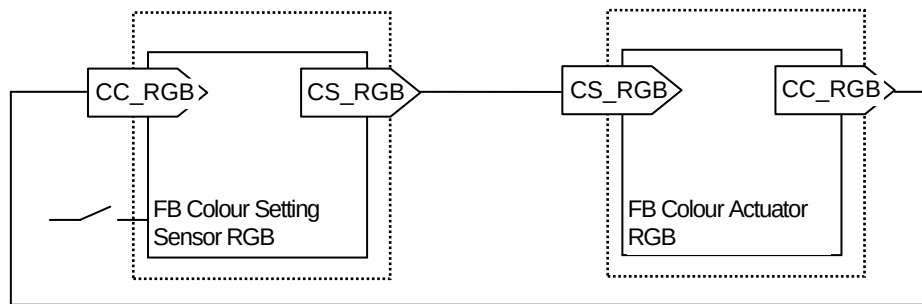


Figure 24 – Basic communication model: RGB Colour Control

Figure 24 gives an example in which the colour of the actuator is controlled from the colour sensor with RGB-Colour Coding.

1.4.4 Constraints

This FB does not foresee any functionality for scene controlling. The Inputs in the colour control actuator for scene control can be controlled by a dedicated FB for scene purposes.

The step sizes for the DP Colour_RC_RGB and DP Colour_RC_RGBW is manufacturer specific. No standard parameter is foreseen to this purpose.

1.4.5 Functional Block Diagram

FB Colour Setting Sensor RGB(W) (CSSRGB)		424
Inputs		Outputs
Combined Info On Off (CIOO)		Combined_Switch On Off (CSOO)
Current Colour RGB (CC_RGB)		Colour Set RGB (CS_RGB)*
Current Colour RGBW (CC_RGBW)		Colour_RC_RGB (RC_RGB)*
		Colour Set RGBW (CS_RGBW)*
		Colour_RC_RGBW (RC_RGBW)*
additional I/Os		Parameters
mandatory One interaction point for triggering transmission of values from output Datapoints.		Absolute RGB Colour Control Value (RGB_CCV)
		Absolute RGBW Colour Control Value (RGBW_CCV)

mandatory optional

* For the allowed combination of output Datapoints see 1.4.7.

Figure 25 – Functional Block Diagram for FB Colour Setting Sensor RGB(W)

1.4.6 Datapoints

Table 8 – Datapoint overview

Datapoint	Description/Remarks	Datapoint Type
Outputs		
Combined Switch On Off	To switch the combined output on or off.	DPT_Switch (1.001)
Colour Set RGB	To control the RGB colour of the lamp	DPT_Colour_RGB (232.600)
Colour_RC_RGB	To relative control the RGB colour of the lamp	DPT_Relative_Control_RGB (254.600)
Colour Set RGBW	To control the RGBW colour of the lamp	DPT_Colour_RGBW (251.600)
Colour_RC_RGBW	To relative control the RGBW colour of the lamp	DPT_Relative_Control_RGBW (252.600)
Inputs		
Combined Info On Off	To reflect the binary status of the combined light output	DPT_Switch (1.001)
Current Colour RGB	To receive the current RGB colour of the lamp	DPT_Colour_RGB (232.600)
Current Colour RGBW	To receive the current RGBW colour of the lamp	DPT_Colour_RGBW (251.600)
Parameters		
Absolute RGB Colour Control Value	Control Value for RGB colour	DPT_Colour_RGB (232.600)
Absolute RGBW Colour Control Value	Control Value for RGBW colour	DPT_Colour_RGBW (251.600)

1.4.7 FB Profiles ⁴⁾

	Basic FB	Standard Mode		LTE-Mode
		FB Profile 1	FB Profile 2	Standard Mode Interface
Features and options				
Input Combined Info On Off	O	(GO)	(GO)	(GO)
Input Current Colour RGB	O	(GO)	(GO)	(GO)
Input Current Colour RGBW	O	(GO)	(GO)	(GO)
Output Combined Switch On Off	O	GO	GO	GO
Output Colour Set RGB	M	GO	GO	GO
Output Colour Set RGBW	O	(GO)	GO	(GO)
Output Colour_RC_RGB	O	(GO)	(GO)	(GO)
Output Colour_RC_RGBW	O	(GO)	(GO)	(GO)

⁴⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

CSSRGB FB Profile 1 models the Colour Setting Sensor RGB(W) without white control; *CSSRGB FB Profile 2* models the Colour Setting Sensor RGB(W) with white control. Table 9 exclusively specifies the allowed combinations of output Datapoints. Combination 1 is for the Colour Setting Control RGB-Sensor, Combination 2 is for the Colour Setting Control RGBW-Sensor.

Table 9 - Combination of output Datapoints in a Colour Setting Sensor RGB(W)

Datapoints	Combination	
	Colour Setting Control RGB-Sensor	Colour Setting Control RGBW-Sensor
	1	2
Combined Switch On Off	M	M
Colour Set RGB	M	M
Colour Set RGBW	O	M
Colour_RC_RGB	O	O
Colour_RC_RGBW	O	O

Table 10 - Parameters

		Basic FB	S-Mode
Parameters	RGB_CCV	O	O
	RGBW_CCV	O	O

As regards the implementation of Parameters in Standard Mode or the LTE-Mode Standard Mode Interface, in case of memory mapped Datapoints the DPT may be implementation specific. In case of implementation as a Property of an Interface Object, the use of the standardised parameter specification is mandatory.

1.4.8 Detailed specification of Datapoints

1.4.9 Output Combined Switch On Off (CSOO)

Name	Combined Switch On Off		CSOO	Mandatory	☒
FB Name:	ColourColour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐
Description					
This Output shall be used to Control the combined info on/off of the colour actuator RGB(W). Transmitting of the value "1" shall cause switching on the combined output, transmitting of the value "0" shall cause switching off the combined output.					
Datapoint Type					
DPT_Name:	DPT_Switch				
DPT Format:	B ₁	DPT_ID:	1.001		
Field	Description	Supp.	Range	Unit	Default
b	Value for the combined light output	M	{0,1}	none	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐ ²⁾	Period:	NO
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Actual value:	☐
Transmit on bus (only for output):			☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

1.4.9.1 Output Colour Set RGB (CS_RGB)

DP Name:	Colour Set RGB	Abbr.:	CS_RGB	Mandatory	☐ ^{a)}
FB Name:	Colour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐
Description					
This Output shall be used to Control the RGB colour of the colour actuator.					
Datapoint Type					
DPT Name:	DPT_Colour_RGB				
DPT Format:	U ₈ U ₈ U ₈	DPT ID:	232.600		
Field	Description	Supp.	Range	Unit	Default
R	This shall be the part of red in the colour set value.	M	0 to 255	none	none
G	This shall be the part of green in the colour set value.	M	0 to 255	none	none
B	This shall be the part of blue in the colour set value.	M	0 to 255	none	none
Access Type					
◆ Output					
☐ this → M	☒	this → 1	☐		
☐ Spontaneous	☒	COV:	☐	Δ-Value:	
☐ Request	☐	Cyclic	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
☐ Power down:	Save:	☐			
☐ Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
The used DPT does not allow indicating either field as invalid or don't care. It is not possible to control only one or two colour primitives and leave the two or one remaining colour primitives unchanged. All fields in the used DPT are always valid. If one of the colour components is not valid, the information cannot be transmitted with this Datapoint (value=0 means: the colour component is switched off, but valid).					
Special Features					
Because of the device dependent interpretation of RGB, this Datapoint is more suited for point-to-point communication, this is, if there is only a single receiver, than for point-to-multipoint communication.					

^{a)} For allowed combinations of output Datapoints see Table 9.

1.4.9.2 Output Colour_Relative Control_RGB (RC_RGB)

DP Name:	Colour Relative Control RGB	Abbr.:	RC_RGB	Mandatory	☐ a)
FB Name:	Colour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐
Description					
This Output shall be used to relatively increment and decrement (StepCodes ≠ 0) the contribution of each colour component (StepCodes ≠ 0) of the colour actuator or to stop the ongoing change (StepCodes = 0). According the requirements concerning the support of ranges by Output DPTs as specified in Chapter 3/7/1 "Interworking Model" clauses 2.1.7.2.1.1 and 2.4 it is allowed that the ranges of the StepCodes be limited to values smaller than 111b. The values 000b shall however always be supported.					
Datapoint Type					
DPT_Name:	DPT_Relative_Control_RGB				
DPT_Format:	B ₁ U ₃ B ₁ U ₃ B ₁ U ₃	DPT_ID:	254.600		
Field	Description	Supp.	Range	Unit	Default
C _R	Shall specify whether the colour component red shall be increased or decreased.	M	{0,1}	none	none
Step Code Colour Red	Shall specify the fading step size or to the code to stop the fading of the colour red.	M	000b to 111b	none	none
C _G	Shall specify whether the colour component green shall be increased or decreased.	M	{0,1}	none	none
Step Code Colour Green	Shall specify the fading step size or to the code to stop the fading of the colour green.	M	000b to 111b	none	none
C _B	Shall specify whether the colour component blue shall be increased or decreased.	M	{0,1}	none	none
Step Code Colour Blue	Shall specify the fading step size or to the code to stop the fading of the colour blue.	M	000b to 111b	none	none
Access Type					
◆ Output					
☐ this → M ☐ Spontaneous ☐ Request	☒ this → 1 ☐ COV: ☐ Cyclic	☐ Δ-Value: ☐ Period:	☐ Min repetition time:		
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

a) For allowed combinations of output Datapoints see Table 9.

1.4.9.3 Output Colour Set RGBW (CS_RGBW)

DP Name:	Colour Set RGBW	Abbr.:	CS_RGBW	Mandatory	☐ a)
FB Name:	Colour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐
Description					
This Output shall be used to Control the RGBW colour of the colour actuator.					
NOTE 2 Opposite to the DPT_Colour_RGB used in the Output Colour Set RGB, the DPT_Colour_RGBW used in this DP foresees mask bits for each colour component. This makes that a sensor can control any colour component separately without influencing the other colour components.					
Datapoint Type					
DPT Name:	DPT_Colour_RGBW				
DPT Format:	U ₈ U ₈ U ₈ U ₈ r ₈ r ₄ B ₄		DPT ID:	251.600	
Field	Description	Supp.	Range	Unit	Default
R	Colour Level Red	M	0 % to 100 %	-	-
G	Colour Level Green	M	0 % to 100 %	-	-
B	Colour Level Blue	M	0 % to 100 %	-	-
W	Colour Level White	M	0 % to 100 %	-	-
m _R	Shall specify whether the colour information red in the field R is valid or not.	M	{0,1}	None.	None.
m _G	Shall specify whether the colour information green in the field G is valid or not.	M	{0,1}	None.	None.
m _B	Shall specify whether the colour information blue in the field B is valid or not.	M	{0,1}	None.	None.
m _W	Shall specify whether the colour information white in the field W is valid or not.	M	{0,1}	None.	None.
Access Type					
◆ Output					
	this → M	☐	this → 1	☒	
	Spontaneous	☒	COV:	☐	Δ-Value:
			Cyclic	☐	Period:
	Request	☐			
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):		☐	Read from bus (only for input):
Exception Handling					
None.					
Special Features					
Because of the device dependent interpretation of RGB, this Datapoint is more suited for point-to-point communication, this is, if there is only a single receiver, than for point-to-multipoint communication.					

a) For allowed combinations of output Datapoints see Table 9.

1.4.9.4 Output Colour Relative Control RGBW (RC_RGBW)

DP Name:	Colour Relative Control RGBW	Abbr.:	RC_RGBW	Mandatory	☐ a)
FB Name:	Colour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐
Description					
This Output shall be used to relatively increment and decrement (StepCodes ≠ 0) the contribution of each colour component (StepCodes ≠ 0) of the colour actuator or to stop the ongoing change (StepCodes = 0). According the requirements concerning the support of ranges by Output DPTs as specified in Chapter 3/7/1 "Interworking Model" clauses 2.1.7.2.1.1 and 2.4 it is allowed that the ranges of the StepCodes be limited to values smaller than 111b. The values 000b shall however always be supported.					
Datapoint Type					
DPT_Name:	DPT_Relative_Control_RGBW				
DPT_Format:	r4B1U3r4B1U3r4B1U3B8		DPT_ID:	252.600	
Field	Description	Supp.	Range	Unit	Default
C _R	Shall specify whether the colour component red shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Colour Red	Shall specify the fading step size or to the code to stop the fading of the colour red	M	000b to 111b	None.	None.
C _G	Shall specify whether the colour component gree shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Colour Green	Shall specify the fading step size or to the code to stop the fading of the colour green	M	000b to 111b	None.	None.
C _B	Shall specify whether the colour component blue shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Colour Blue	Shall specify the fading step size or to the code to stop the fading of the colour blue	M	000b to 111b	None.	None.
C _W	Shall specify whether the colour component blue shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Colour White	Shall specify the fading step size or to the code to stop the fading of the colour white	M	000b to 111b	None.	None.
Masks: b ₃	This field shall indicate the validity of the fields C _R and Step Code Colour Red.	M	{0,1}	None.	None.
Masks: b ₂	This field shall indicate the validity of the fields C _G and Step Code Colour Green.	M	{0,1}	None.	None.
Masks: b ₁	This field shall indicate the validity of the fields C _B and Step Code Colour Blue.	M	{0,1}	None.	None.
Masks: b ₀	This field shall indicate the validity of the fields C _W and Step Code Colour White.	M	{0,1}	None.	None.
Access Type					
◆ Output					
this → M	☐	this → 1	☒		
Spontaneous	☒	COV:	☐	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐

DP Name:	Colour Relative Control RGBW	Abbr.:	RC_RGBW	Mandatory	☐ a)
FB Name:	Colour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐
Exception Handling					
None.					
Special Features					
None.					

a) For allowed combinations of output Datapoints see Table 9.

1.4.9.5 Input Current Colour RGB (CC_RGB)

DP Name:	Actual Colour RGB (CC_RGB)		CC_RGB	Mandatory	☐			
FB Name:	Colour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐			
Description								
This Input shall be used to receive the RGB colour of the colour actuator.								
Datapoint Type								
DPT_Name:	DPT_Colour_RGB							
DPT_Format:	U ₈ U ₈ U ₈	DPT_ID:	232.600					
Field	Description	Supp.	Range	Unit	Default			
R	This shall be the part of red in the colour set value.	M	0 to 255	none	none			
G	This shall be the part of green in the colour set value.	M	0 to 255	none	none			
B	This shall be the part of blue in the colour set value.	M	0 to 255	none	none			
Access Type								
◆ Input								
N → this	☐	1 → this	☒					
Spontaneous Request	☒	Cyclically:	☐	Time-out:				
	☐	Polling:	☐	Period:				
Communication Type								
◆ Group Object Datapoint				Mandatory:	☒			
Default Group Address:	---							
Dynamics								
Power down:	Save:	☐						
Power up:	Value:	No initialisation:	☒	Default value:	☐			
		Saved value:	☐	Actual value:	☐			
Transmit on bus (only for output):		☐	Read from bus (only for input):					
		☐						
Exception Handling								
If one of the colour components is not valid, the information cannot be transmitted with this datapoint (value=0 means: the colour component is switched off, but valid)								
Special Features								
None.								

1.4.9.6 Input Current Colour RGBW (CC_RGBW)

DP Name:	Actual Colour_RGBW (CC_RGBW)		CC_RGBW	Mandatory	☐
FB Name:	Colour Setting Sensor RGB(W) (CSSRGB)			Can be internal	☐
Description					
This Input shall be used to receive the RGBW colour of the colour actuator.					
Datapoint Type					
DPT_Name:	DPT_Colour_RGBW				
DPT Format:	U ₈ U ₈ U ₈ U ₈ r ₄ B ₄	DPT_ID:	251.600		
Field	Description	Supp.	Range	Unit	Default
R	Colour Level Red	M	0-255	-	-
G	Colour Level Green	M	0-255	-	-
B	Colour Level Blue	M	0-255	-	-
W	Colour Level White	M	0-255	-	-
m _R	Shall specify whether the colour information red in the field R is valid or not.	M	{0,1}	-	-
m _G	Shall specify whether the colour information green in the field G is valid or not.	M	{0,1}	-	-
m _B	Shall specify whether the colour information blue in the field B is valid or not.	M	{0,1}	-	-
m _W	Shall specify whether the colour information white in the field W is valid or not.	M	{0,1}	-	-
Access Type					
◆ Input					
N → this	☐	1 → this	☒		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Actual value:	☐
Transmit on bus (only for output):			☐	Read from bus (only for input): ☐	
Exception Handling					
Special Features					
None.					

1.4.9.7 Parameter Absolute RGB Colour Control Value (RGB_CCV)

FB:	Colour Setting Sensor RGB(W) (CSSRGB)	Property Name (<u>Server</u>):		Absolute_ RGB_Control_ Value (RGB_CCV)	Mandatory ☐ Optional ☒
Description:					
This parameter shall define the value that shall be sent with the output Colour Set RGB (CS_RGB).					
DPT:	Name	DPT_Colour_- RGB	DPT ID	232.600	Datatype format U ₈ U ₈ U ₈
Field	Description			Sup.	Range
R	This shall be the part of red in the colour set value.			M	0 to 255
G	This shall be the part of green in the colour set value.			M	0 to 255
B	This shall be the part of blue in the colour set value.			M	0 to 255
Communication:					
DP Address:	object_type:	424	PID:	101	
(in the server)	start_index:	1	nr_of_elem:	1	
Property access:	Read only ☐	Read/Write ☒			
Protection	Read level	-	Write level	-	
Exception Handling:	Value after Power-up:	Stored Value ☒	Act Value ☐	Default Value ☐	
None.					
Special Features:					
None.					

1.4.9.8 Parameter Absolute RGBW Colour Control Value (RGBW_CCV)

FB:	Colour Setting Sensor RGB(W) (CSSRGB)	Property Name (Server):		Absolute_ RGBW_Contro l_Value (RGBW_CCV)	Mandatory ☐ Optional ☒
Description:					
This parameter shall define the value that shall be sent with the output Colour Set RGBW (CS_RGBW).					
DPT:	Name	DPT_Colour_- RGBW	DPT ID	251.600	Datatype format U8U8U8r4B4
Field	Description			Sup.	Range
R	This shall be the part of red in the colour set value.			M	0 to 255
G	This shall be the part of green in the colour set value.			M	0 to 255
B	This shall be the part of blue in the colour set value.			M	0 to 255
W	This shall be the part of white in the colour set value.			M	0 to 255
m _R	Shall specify whether the colour information red in the field R is valid or not.			M	{0,1}
m _G	Shall specify whether the colour information green in the field G is valid or not.			M	{0,1}
m _B	Shall specify whether the colour information blue in the field B is valid or not.			M	{0,1}
m _W	Shall specify whether the colour information white in the field W is valid or not.			M	{0,1}
Communication:					
DP Address:	object_type:	424	PID:	102	
(in the server)	start_index:	1	nr_of_elem:	1	
Property access:	Read only ☐	Read/Write ☒			
Protection	Read level	-	Write level	-	
Exception Handling:	Value after Power-up:	Stored Value ☒	Act Value ☐	Default Value ☐	
None.					
Special Features:					
None.					

1.5 FB Colour Setting Sensor xyY (Object Type 425)

1.5.1 Aims and objectives

1.5.1.1 General

The control of the xyY colour considers the possibility to generate professional colour control, mostly for commercial applications (e.g. in a museum, shops for meat, vegetables, fruit ...) or for other applications where exact colour referenced luminaires are necessary.

1.5.1.2 Introduction and overview

The FB Colour Setting Sensor xyY is used in the Application Domain Lighting for providing input data to a colour actuator. It specifies the functionality, for example contained in a room control unit (e.g. with display for change of the colour with dedicated user interaction), to modify the colour of a lamp.

The implementation of this functionality is not restricted to push buttons only. It may for instance be implemented in a larger controller, scheduler, gateway ...

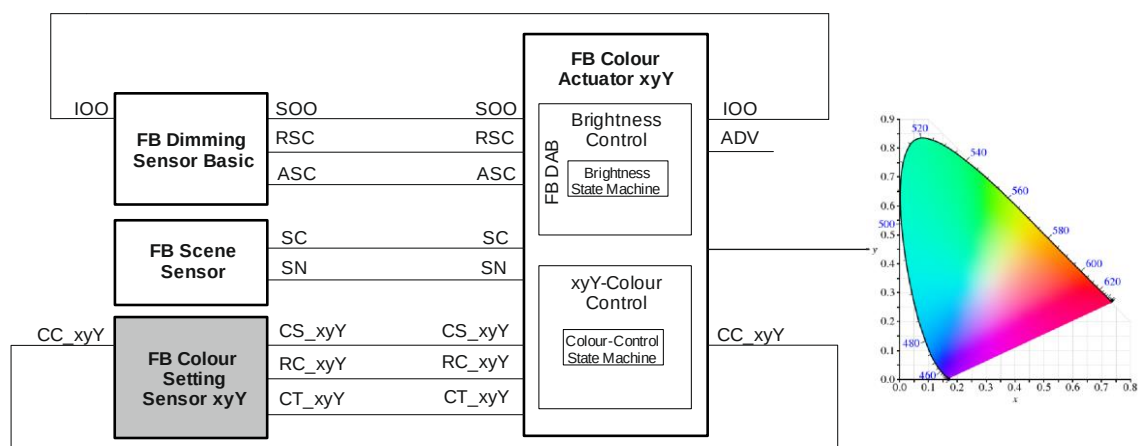


Figure 26 – Overview of the Application Model Colour Control in the Application Domain Lighting

Respective to different colour coding models the FB shall at least support the following model:

- xyY

Other encodings and models may additionally be supported.

1.5.1.3 Colour model xyY

The colour triangle is a plane figure scaled in such a way that any colour within it may be identified with two coordinates x and y , where x and y are both in a range 0 to 1. The total output light intensity depends on the requested xy -coordinate and the brightness (luminance) Y .

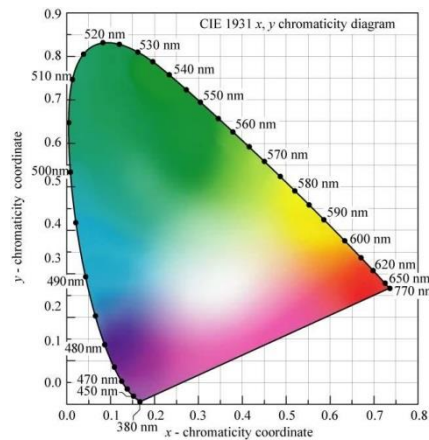


Figure 27 - Control of colour with Colour Type xyY

Third dimension: brightness or luminance (indicated by Y).

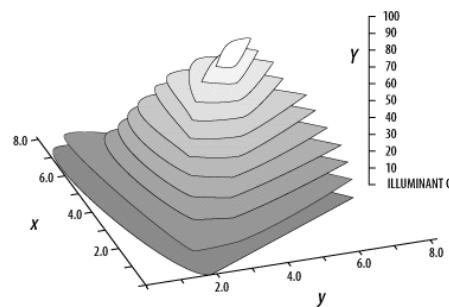


Figure 28 – xyY diagram

Using the xyY values, any two colours can be compared to determine whether they match. This is the whole purpose of CIE's standards.

Summary for the use of xyY control

It is useful for the more professional applications. As this interpretation is device independent, it is better suited in communication models with multiple receivers.

For colour control within the xyY -Colour Model, it shall optionally be possible to control the colour relative.

- EXAMPLE 2 Curve A shows the change of colour while the saturation is kept constant (distance to the White Point WP)
- EXAMPLE 3 Curve B shows the change of saturation (distance to the White Point) while the colour is kept constant
- EXAMPLE 4 Curve C shows the simultaneous change of saturation and colour
- EXAMPLE 5 Not shown here is the simultaneous change of Brightness.

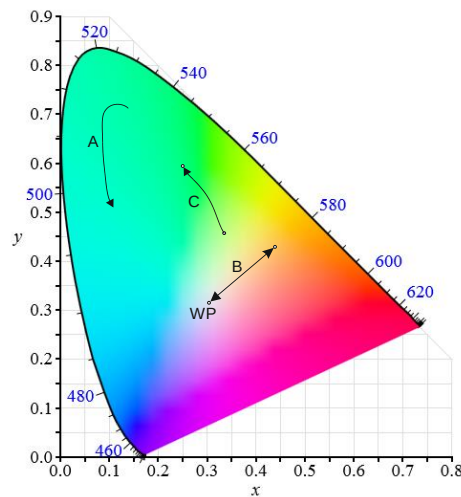


Figure 29 – CIE x,y chromatic diagram – relative colour control

1.5.1.4 Various

Display elements (with own FBs) can be integrated to show the status of the colour control actuator.

The Inputs and Outputs of the Functional Block are described, but not the Human Machine Interface (HMI). Consequently, the manufacturers of the colour control sensor devices have the opportunity to implement their design and their operation methods.

1.5.2 Functional specification

1.5.2.1 Overview

The FB Colour Setting Sensor xyY provides hardwired inputs for triggering transmission of values of output Datapoints.

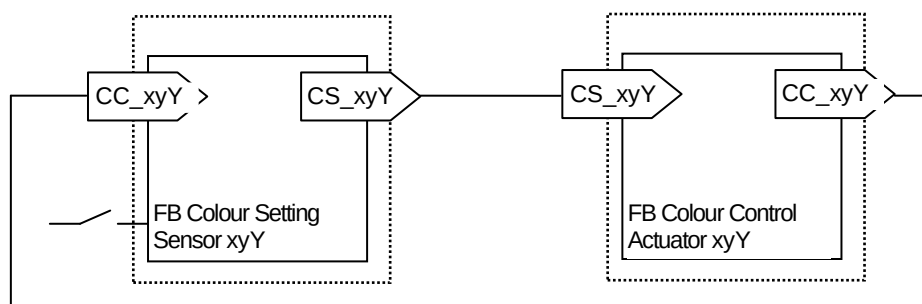


Figure 30 – Basic communication model: xyY Colour Control

Figure 30 gives an example in which the colour of the actuator is controlled from the colour setting sensor with xyY colour Coding.

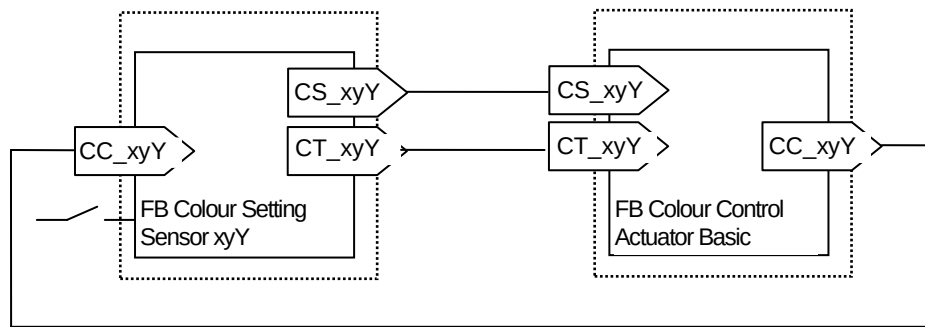


Figure 31 – communication model: Colour Control Sensor with Colour Control Actuator

Figure 31 gives an example in which the colour is controlled from the colour setting sensor with xyY-Colour Coding (DP CS_xyY). In addition it should be possible to control the xyY-Colour transition for dynamic aspects (DP CT_xyY).

1.5.3 Constraints

This FB does not foresee any functionality for scene controlling. The Inputs in the colour control actuator for scene control can be controlled by a dedicated FB for scene purposes.

The step sizes for the DP Colour_RC_RGB and DP Colour_RC_RGBW are implementation specific. No standard parameter is foreseen to this purpose.

1.5.4 Functional Block Diagram

FB Colour Setting Sensor xyY (CSSXY)		425
Inputs		Outputs
Current Colour xyY (CC_xyY)		Colour_Set_xyY (CS_xyY)*
		Colour_RC_xyY (RC_xyY)
		Colour_Transition_xyY (CT_xyY)*
additional I/Os		Parameters
mandatory One interaction point for triggering transmission of values from output Datapoints.		Absolute xyY Colour Control Value (XYX_CCV)
		Absolute xyY Colour Transition Time (XYX_CCTT)

mandatory optional

* For the allowed combination of output Datapoints see 1.5.6.

Figure 32 – Functional Block Diagram for FB Colour Setting Sensor xyY

1.5.5 Datapoints

Table 11 – Datapoint overview

Datapoint	Description/Remarks	Datapoint Type
Outputs		
Colour_Set_xyY	To control the xyY colour of the lamp	DPT_Colour_xyY (242.600)
Colour_RC_xyY	To relative control the xyY colour of the lamp	DPT_Relative_Control_xyY (253.600)
Colour_Transition_xyY	To control the xyY colour transition of the lamp	DPT_Colour_Transition_xyY (243.600)

Datapoint	Description/Remarks	Datapoint Type
Inputs		
Current Colour xyY	To receive the current xyY colour of the lamp	DPT_Colour_xyY (242.600)
Parameters		
Absolute xyY Colour Control Value	Control Value for xyY colour	DPT_Colour_xyY (242.600)
Absolute xyY Colour Transition Time	Transition Time for xyY colour transition	DPT_TimePeriod100-Msec (7.004)

1.5.6 FB Profiles ⁵⁾

Features and options	Basic FB	Standard Mode	LTE-Mode
		FB Profile 1	Standard Mode Interface
Input Current Colour xyY	O	(GO)	(GO)
Output Colour Set xyY	O	GO	GO
Output Colour_RC_xyY	O	(GO)	(GO)
Output Colour_Transition_xyY	O	(GO)	(GO)

CSSxyY FB Profile 1 models the Colour Setting Sensor xyY without any additional Inputs or Outputs. This is the minimal requirement.

This can optionally be extended with the Outputs for relative xyY colour control (Output Colour_RC_xyY) and xyY colour transition (Output Colour_Transition_xyY). Further, the Input Current Colour xyY may be added.

In case of Profile 1 it may also be possible to optionally integrate (beside DP Colour_RC_xyY) in addition three separate, additional Datapoints according DPT_Control_Dimming(3.007) for separate relative control of the xyY-colour (one Datapoint for relative control of brightness, one Datapoint for relative control of colour and one Datapoint for relative control of saturation).

Table 12 - Parameters

		Basic FB	S-Mode
Parameters	XYX_CCV	O	O
	XYX_CCTT	O	O

As regards the implementation of Parameters in Standard Mode or the LTE-Mode Standard Mode Interface, in case of memory mapped Datapoints the DPT may be implementation specific. In case of implementation as a Property of an Interface Object, the use of the standardised parameter specification is mandatory.

⁵⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

1.5.7 Detailed specification of Datapoints

1.5.7.1 Output Colour_Set_xyY (CS_xyY)

DP Name:	Colour_Set_xyY	Abbr.:	CS_xyY	Mandatory	☐ a)
FB Name:	Colour Setting Sensor xyY (CSSXYY)			Can be internal	☐
Description					
This Output shall be used to Control the xyY colour of the colour actuator.					
Datapoint Type					
DPT Name:	DPT Colour_xyY				
DPT Format:	U ₁₆ U ₁₆ U ₈ r ₆ B ₂	DPT ID:	242.600		
Field	Description	Supp.	Range	Unit	Default
x-axis	x-coordinate of the colour information	M	0 to 65 535	None.	None.
y-axis	y-coordinate of the colour information	M	0 to 65 535	None.	None.
Brightness	Brightness of the colour	M	0 % to 100 %	None.	None.
C	This field shall indicate whether the colour information in the fields x-axis and y-axis is valid or not.	M	{0,1}	None.	None.
B	This field shall indicate whether the brightness information in the field <i>Brightness</i> is valid or not.	M	{0,1}	None.	None.
Access Type					
◆ Output					
☐	this → M	☒	this → 1	☐	
☐	Spontaneous	☒	COV:	☐	Δ-Value: Min repetition time:
☐			Cyclic	☐	Period:
☐	Request	☐			
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
☐	Power down:	Save:	☐		
☐	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):		☐	Read from bus (only for input):
Exception Handling					
None.					
Special Features					
None.					

a) For allowed combinations of output Datapoints see 1.5.6.

1.5.7.2 Output Colour_Relative Control_xyY (RC_xyY)

DP Name:	Colour Relative Control xyY	Abbr.:	RC_xyY	Mandatory	☐ a)
FB Name:	Colour Setting Sensor xyY (CSSXY)	Can be internal			☐
Description					
This Output shall be used to simultaneous increase or decrease the brightness, the colour and the saturation (StepCodes ≠ 0) of the colour actuator or to stop the ongoing change (StepCodes = 0). According the requirements concerning the support of ranges by Output DPTs as specified in Chapter 3/7/1 "Interworking Model" clauses 2.1.7.2.1.1 and 2.4 it is allowed that the ranges of the StepCodes be limited to values smaller than 111b. The values 000b shall however always be supported.					
Datapoint Type					
DPT_Name:	DPT_Relative_Control_xyY				
DPT_Format:	r ₄ B ₁ U ₃ r ₄ B ₁ U ₃ r ₄ B ₁ U ₃ B ₈	DPT_ID:	253.600		
Field	Description	Supp.	Range	Unit	Default
c _s	Shall specify whether the saturation of the colour shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Saturation	Shall specify the fading step size or to the code to stop the fading of the saturation	M	000b to 111b	None.	None.
c _c	Shall specify whether the colour shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Colour	Shall specify the fading step size or to the code to stop the fading of the colour	M	000b to 111b	None.	None.
c _B	Shall specify whether the brightness shall be increased or decreased.	M	{0,1}	None.	None.
Step Code Brightness	Shall specify the fading step size or to the code to stop the fading of the brightness	M	000b to 111b	None.	None.
Masks: b ₂	This field shall indicate whether the fields c _s and Step Code Saturation are valid or not.	M	{0,1}	None.	None.
Masks: b ₁	This field shall indicate whether the fields c _c and Step Code Colour are valid or not.	M	{0,1}	None.	None.
Masks: b ₀	This field shall indicate whether the fields c _B and Step Code Brightness are valid or not.	M	{0,1}	None.	None.
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☐	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	---				
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

a) For allowed combinations of output Datapoints see 1.5.6.

1.5.7.3 Output Colour_Transition_xyY (CT_xyY)

DP Name:	Colour_Transition_xyY	Abbr.:	CT_xyY	Mandatory	☐ ^{a)}
FB Name:	Colour Setting Sensor xyY (CSSXY)			Can be internal	☐
Description					
This Output shall be used to Control the transition for the xyY colour of the colour actuator.					
Datapoint Type					
DPT_Name:	DPT_Colour_Transition_xyY				
DPT_Format:	U ₁₆ U ₁₆ U ₁₆ U ₈ f ₆ B ₂		DPT_ID:	243.600	
Field	Description	Supp.	Range	Unit	Default
Time Period	Unsigned time-value for calculating fading time.	M	0s to 6553,5 s	100 ms	none
x-axis	x-coordinate of the colour information	M	0 to 65 535	none	none
y-axis	y-coordinate of the colour information	M	0 to 65 535	none	none
Brightness	Brightness of the colour	M	0 % to 100 %	none	none
C	This field shall indicate whether the colour information in the fields x-axis and y-axis is valid or not.	M	{0,1}	none	none
B	This field shall indicate whether the brightness information in the field <i>Brightness</i> is valid or not.	M	{0,1}	none	none
Access Type					
◆ Output					
☐ this → M ☐ Spontaneous ☐ Request	☒ this → 1 ☐ COV: ☐ Cyclic	☐ Δ-Value: ☐ Period:	Min repetition time:		
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

^{a)} For allowed combinations of output Datapoints see 1.5.6.

1.5.7.4 Input Current Colour xyY (CC_xyY)

DP Name:	Current Colour_xyY		CC_xyY	Mandatory	☐
FB Name:	Colour Setting Sensor xyY (CSSXY)			Can be internal	☐
Description					
This Input shall be used to receive the xyY colour of the colour actuator.					
Datapoint Type					
DPT_Name:	DPT_Colour_xyY				
DPT Format:	U ₁₆ U ₁₆ U ₈ R ₆ B ₂		DPT_ID:	242.600	
Field	Description	Supp.	Range	Unit	Default
x-axis	x-coordinate of the colour information	M	0 to65 535	None.	None.
y-axis	y-coordinate of the colour information	M	0 to65 535	None.	None.
Brightness	Brightness of the colour	M	0 % to 100 %	None.	None.
C	This field shall indicate whether the colour information in the fields x-axis and y-axis is valid or not.	M	{0,1}	None.	None.
B	This field shall indicate whether the brightness information in the field <i>Brightness</i> is valid or not.	M	{0,1}	None.	None.
Access Type					
◆ Input					
	N → this	☐	1 → this	☒	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
	Default Group Address:	---			
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
		Saved value:	☐	Actual value:	☐
	Transmit on bus (only for output):			☐	Read from bus (only for input):
Exception Handling					
None.					
Special Features					
None.					

1.5.7.5 Parameter Absolute xyY Colour Control Value (XYY_CCV)

FB:	Colour Setting Sensor xyY (CSSXYY)	Property Name (Server):	Absolute_xyY_Control_Value (XYY_CCV)	Mandatory ☐	Optional ☒
Description:					
This parameter shall define the value that shall be sent with the output Colour_Set_xyY (CS_xyY).					
DPT:	Name	DPT_Colour_xyY	DPT ID	242.600	Datatype format U ₁₆ U ₁₆ U ₈ r ₆ B ₂
Field	Description		Sup.	Range	Unit Default
x-axis	x-coordinate of the colour information		M	0-65535	None . None.
y-axis	y-coordinate of the colour information		M	0-65535	None . None.
Brightness	Brightness of the colour		M	0 % to 100 %	% None.
C	This field shall indicate whether the colour information in the fields <i>x-axis</i> and <i>y-axis</i> is valid or not.		M	{0,1}	None . None.
B	This field shall indicate whether the brightness information in the field <i>Brightness</i> is valid or not.		M	{0,1}	None . None.
Communication:					
DP Address:	object_type:	425	PID:	101	
(in the server)	start_index:	1	nr_of_elem:	1	
Property access:	Read only ☐	Read/Write ☒			
Protection	Read level	-	Write level	-	
Exception Handling:	Value after Power-up:	Stored Value ☒	Act Value ☐	Default Value ☐	
None.					
Special Features:					
None.					

1.5.7.6 Parameter Absolute xyY Colour Transition Time (XYY_CCTT)

FB:	Colour Setting Sensor xyY (CSSXYY)	Property Name (Server):	Absolute_xyY_Transition_Time (XYY_CCTT)	Mandatory ☐	Optional ☒
Description:					
This parameter shall define the transition time that shall be sent with the output Colour_Transition_xyY (CT_xyY).					
DPT:	Name	DPT_TimePeriod100Msec	DPT ID	7.004	Datatype format U ₁₆
Field	Description		Sup.	Range	Unit Default
Time Period	Unsigned time-value for calculating fading time.		1 ms to 65 535 ms	100 ms	10 ms None.
Communication:					
DP Address:	object_type:	425	PID:	102	
(in the server)	start_index:	1	nr_of_elem:	1	
Property access:	Read only ☐	Read/Write ☒			
Protection	Read level	-	Write level	-	
Exception Handling:	Value after Power-up:	Stored Value ☒	Act Value ☐	Default Value ☐	
None.					
Special Features:					
None.					

2 FB Movement Detector for Lighting

2.1 Aims and objectives

The “Movement Detector for Lighting“ (FB MDL) shall control actuators for lighting in function of:

- the detection of movement, and
- the detected brightness.

The brightness information may be retrieved either from a hardwired input, or from the Input Brightness External, e.g. provided by the FB Room Light Sensor.

The dependency of the functionality on this brightness is the basic difference between this FB MDL and similar FBs for HVAC and for Anti-Intrusion.

NOTE 3 This FB Movement Detector for Lighting is primarily designed for use in the application domain Lighting. Its 1 bit Outputs however allow this FB to be used in other application domains as well. Further integration in other application domains is possible through generic FBs, as FB Scene Control. See 2.2.11.

2.2 Functional specification

2.2.1 Basic functionality

The FB MDL shall be able to detect movement in a physical movement sensor, typically PIR (Passive InfrRed), and to transmit this information on the bus via at least one Output Switch On Off (SOO) or Timed StartStop (TSS).

The FB MDL may have further Inputs and functionality.

1. Input Movement Trigger shall serve for receiving movement signals from an external movement detector.
2. Input Info OnOff shall serve to suppress the faulty detection of movement at the moment that the light is switched.
3. The brightness evaluation shall serve for suppressing light switching if the brightness in the detection area is above a certain threshold level.

The combination and sequence of these functions is not specified. Further on in this specification multiple use cases are specified. The choice for the design of the realisation of this FB MDL depends on the main use case that is intended to be solved.

Figure 33 gives a possible design; alternative solutions are possible as well.

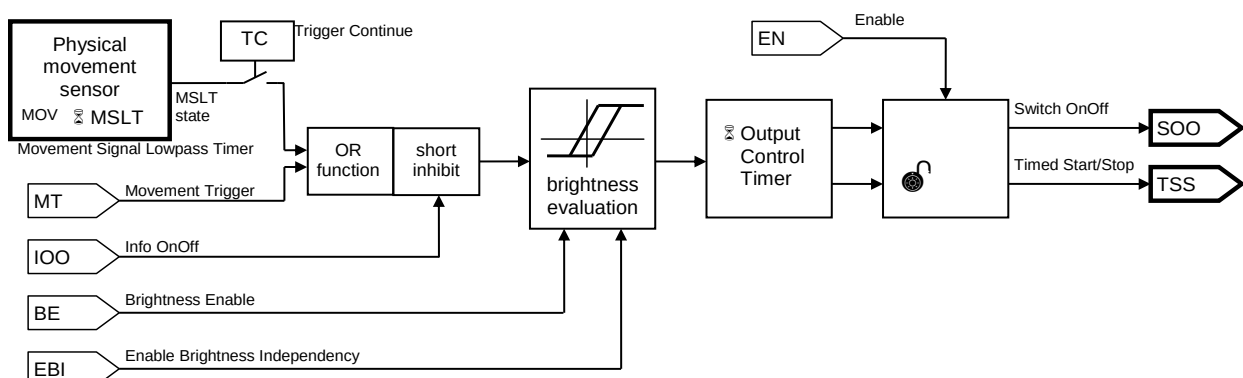


Figure 33 – Possible design of FB Movement Detector for Lighting

2.2.2 Physical means to detect movement

The FB MDL shall have a physical movement sensor to be able to detect movement. This can be done through the use of various physical means, such as PIR, microwaves ... These physical means may also be combined. These means are however not modelled differently in this KNX FB MDL. All movement detectors for lighting are modelled through this one single common FB MDL without taking the physical detection method into consideration any further.

A first timer in the FB MDL shall be the Movement Signal Lowpass Timer (MSLT).

Timer MSLT is started and retriggered on any physical detection of movement. A running timer MSLT thus signals that the device has detected a valid movement signal.

Timer MSLT is the timer that shall prevent that if movement is detected with a higher frequency (e.g. multiple detections during 1 second) that this leads to as many transmissions on the bus.

The timer MSLT depends on the applied hardware, electronics, etc. and is product specific. Therefore, the timer MSLT is not available as an application parameter.

The state of the MSLT timer shall be the output signal of this physical movement sensor.

2.2.3 Outputs *Switch OnOff* and *Timed StartStop*

These Outputs reflect two typical use cases of this FB MDL.

Table 13 – Comparison SOO to TSS

Event	Use case 1	Use case 2
	SOO shall	TSS shall
• first detection of movement	transmit the value On	transmit the value Start.
• detected movement continues	optionally repeat its value.	repeat its value periodically.
• no more movement is detected	transmit the value Off	not transmit the value Stop

2.2.4 Time dependencies

2.2.4.1 Timer definitions

Next to the above specified timer MSLT, FB MDL shall have a further timer. This second timer shall be the Output Control Timer (OCT).

Timer OCT shall be the timer that shall control

- the transmission of SOO, *or*
- the periodic transmission of TSS.

The time-out value of OCT is available as application parameter.

EXAMPLE 6 Timer MSLT is 5 s to 10 s; Timer OCT is 45 s.

2.2.4.2 For Timed StartStop

Timer MSLT shall be started on the first detection of movement. It shall be retriggered on every renewed detection of movement.

Timer OCT shall be started when Timer MSLT is started. When Timer OCT expires and Timer MSLT is running, then Timer OCT shall be automatically restarted. Timer OCT is never retriggered and never stopped.

Output TSS shall transmit the value Start when Timer OCT is started or restarted. The Output TSS shall in this case never transmit the value Stop.

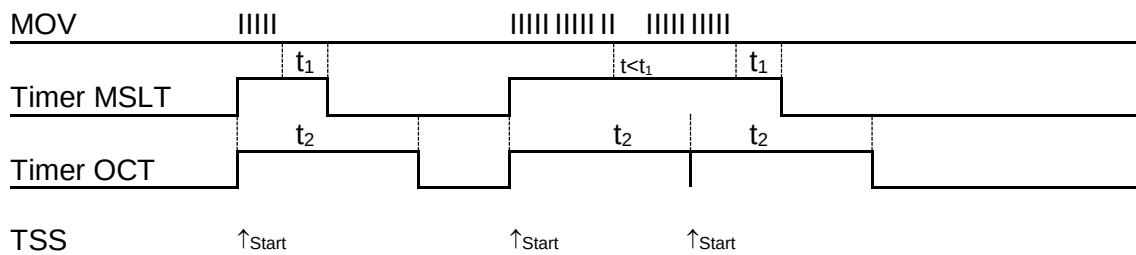


Figure 34 – Functionality of Timed StartStop

2.2.4.3 For Switch OnOff

Timer MSLT shall be started on the first detection of movement. It shall be retriggered on every renewed detection of motion.

Timer OCT shall be reset and started if Timer MSLT expires. Timer OCT is stopped if Timer MSLT is started.

Output SOO shall transmit the value On when Timer MSLT is started and Timer OCT is not running.

Output SOO shall transmit the value Off if Timer OCT expires. It is not transmitted if Timer MSLT expires or if Timer OCT is stopped.

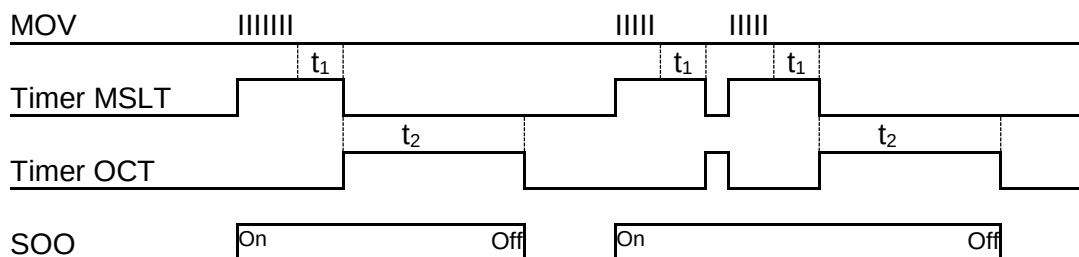


Figure 35 – Functionality of Switch OnOff

2.2.5 Brightness dependency

The functionality of the FB MDL shall further be influenced by the current brightness.

USE CASE 3 If movement is detected, the FB MDL will switch on the light if the brightness is below the threshold. The measured brightness will by this increase. Yet, this FB MDL should again switch off the light when it no longer detects any movement. This means that while OCT runs, the FB MDL works independently of the measured brightness.

If a first movement is detected, the FB MDL will switch on the light, via SOO or TSS. If further movement is detected, SOO may and TSS shall be repeated. If no more movement is detected then SOO shall transmit the value Off. This means that the brightness is only evaluated for the initial movement detection when the timers MSLT and OCT are started.

1. The initial reaction of the FB MDL, this is, on the first detection of movement, when both timers are expired, shall be as specified in Table 14.

Table 14 – INITIAL reaction of FB MDL

Current brightness	SOO	TSS
below threshold value	Shall transmit On.	Shall transmit Start.
above threshold value	No action.	No action.

2. When either one of the timers Timer MSLT or Timer OCT is running, the functionality of the FB MDL shall be independent of the measured brightness and shall be as specified in 2.2.4.

The Parameter Enable Brightness Independency (EBI) shall be used to control this brightness dependency. If EBI is not implemented, then the FB MDL shall always operate brightness dependent concerning the initial detection of movement.

The threshold value may optionally be controlled through the Parameter Brightness Threshold as specified in 2.7.7.

The current brightness is typically obtained from a built-in brightness sensor. It may as well (additionally) be received through the Input Brightness External. In case a built-in brightness sensor is combined with one or more external brightness sensors, the relation between both is implementation dependent.

2.2.6 Input Info OnOff and Parameter Movement Sensor Pause Time

The Input “Info OnOff” shall be used to receive the status of an actuator.

This shall allow the FB MDL to know the moment when a light is switched, off or on, to suppress a movement detection from the light source.

On reception any value, On or Off, on the Input Info OnOff, the FB MDL shall not evaluate the physical movement signals for a short time. This time can be fixed in the device or can be given by the parameter Movement Sensor Pause Time (MSPT). During this time the physical movement sensor signals shall not be evaluated. The implementation of this is not fixed; it may be implemented for instance as follows (informative).

- On reception of a telegram on the Input Info OnOff, the FB MDL will start a timer, with a value according the parameter MSPT if implemented, during which the internal signal MSLT will not cause the transmission of SOO or TSS.
- On detection of movement through the internal signal MSLT, a timer is started, with a value according the parameter MSPT if implemented. If any telegram is received on the Input IOO while this timer is running then there will be no transmission of SOO or TSS; if no telegram is received on the Input IOO, then SOO and TSS will be handled as normal.
- Combinations of the above and alternative solutions can be considered as well.

EXAMPLE 7 3 s

This Input shall also allow to inhibit the evaluation of the internal signal MSLT state, as explained in USE CASE 6 and in the specification of the parameter Trigger continue in clause 2.2.9.

2.2.7 Input Enable

The Input Enable shall be used to limit the transmission of the Outputs TSS and SOO. The influence of this Input may be located at multiple possible locations inside the FB MDL. Depending on this, the effect of the Input Enable may be

- to stop all communication of TSS and SOO, or
- to stop only the evaluation of the internal movement sensor, or
- to stop the evaluation of the external signal MT,

or any combination of the above.

Input Enable is however not intended to disable the brightness evaluation. To this, the Input Enable Brightness Independency shall be used (see 2.2.5).

USE CASE 4 This functionality can be used to disable the light switching by a movement detector at night e.g. without forcing (Input Forced) an actuator.

USE CASE 5 During cleaning of the area that is covered by this movement detector, this Input allows preventing that the light is switched on and off repeatedly.

2.2.8 Input Movement Trigger

The Input Movement Trigger (MT) serves for connecting external movement detecting.

The evaluation of the Input Movement Trigger (MT) is not fixed.

1. It may be evaluated in the same way as the internal physical movement sensor signal (MSLT state) of the FB MDL. It then starts the timer OCT as specified in 2.2.4. The transmission of the Outputs TSS or SOO may be under the control of the Parameter Movement Trigger Evaluation (see 2.7.10).

USE CASE 6 If the light within a room is controlled through push buttons as well as through movement detectors, it may be wanted that the light is switched on through a push button but is switched off automatically by the movement detector. To allow this, the push button communicates through this Input with the FB MDL instead of directly with the light actuator.

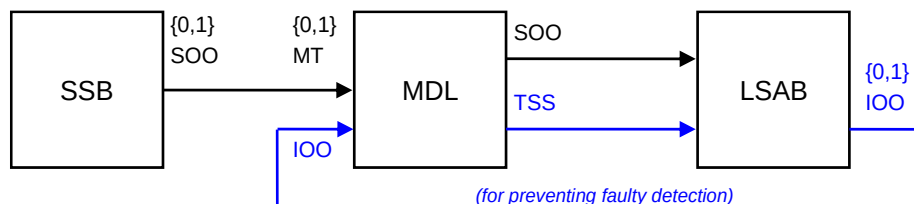


Figure 36 – USE CASE 6 – solution 1
FB MDL evaluates timers and brightness in the interpretation of MT

- In the design of the FB MDL in this solution, the Input MT does not directly affect the Outputs, but is subject to the timer OCT. When this timer expires, it will switch off the light. This model guarantees that the light is always switched off, even if the FB MDL did not detect movement itself.
 - If the design of the FB MDL still evaluates the internal brightness sensor, it may occur that the request for switching on the light from the FB LSSB, is not passed by the FB MDL if the brightness is below the threshold value. The light is not switched on, even though it is requested by a person. This can be solved by the Parameter Movement Trigger Evaluation, which makes that the brightness is not evaluated for the Input MT.
2. The Input MT may also be evaluated elsewhere in the functional scheme of the FB MDL (Figure 33). The Input MT can be implemented so that it directly controls the transmission of the Outputs SOO or TSS, without further evaluation of timers or the brightness. This allows solving USE CASE 6 as in Figure 37.

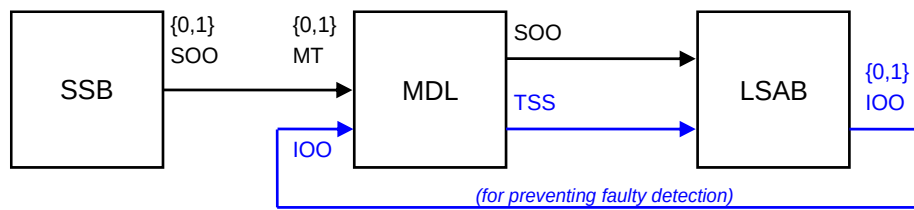


Figure 37 – USE CASE 6 – solution 2
FB MDL interprets MT regardless of timers and brightness

Alternative, more advanced interpretations of MT are possible as well.

NOTE 4 If an implementation of the FB MDL always evaluates the brightness and does not have the Parameter Movement Trigger Evaluation, the functionality can still be reached by having the LSSB and MDL work in parallel.

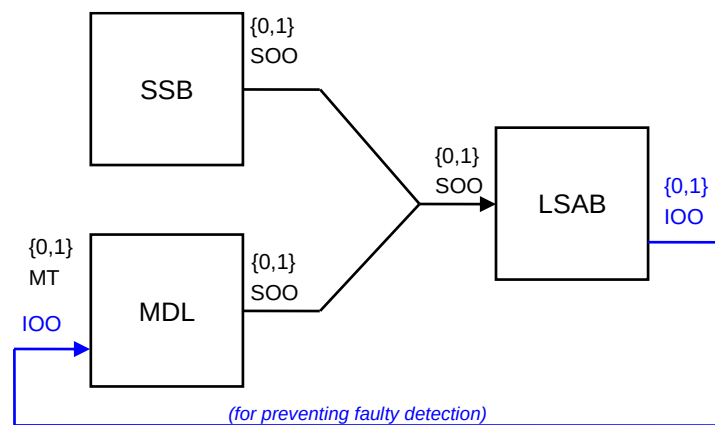


Figure 38 – USE CASE 6 – solution 3
FB MDL evaluates timers and brightness in the interpretation of MT;
LSSB is configured in parallel to MDL

- The FB LSSB can always switch the light, even if the brightness is above the threshold value of the movement sensor.
- The disadvantage of this solution is that if the person only enters the room shortly, or not at all, then the FB MDL has not detected any movement, and will thus not automatically switch off the light.

2.2.9 Trigger continue

If the Parameter Trigger Continue (TC) is set to Enable, then the motion detector shall be active only if the output of the actuator is really switched on. The FB MDL shall recognise this actuator state by its Input Info OnOff. The FB MDL then retriggers the actuator channel while cyclically sending the value On on the Output Switch OnOff.

If the Parameter Trigger Continue is set to Disable then the FB MDL works independently from the Datapoint Info OnOff except the suppression of switching of the light.

2.2.10 Master/slave relation between FB MDL

2.2.10.1 Introduction

If the lighting in a larger area is controlled by more than one movement detector, then it shall be avoided that one movement detector switches the light off, while another movement detector switches the light on. This may be solved by a master/slave concept between the movement detectors. One movement detector controls the lighting; the other movement detector(s) do not control the lighting but report their movement detection through their Outputs TSS to the master via the Input Movement Trigger of the master. The functionality of the master FB MDL is further unchanged, i.e. it is further controlled through its Inputs Enable, etc.

2.2.10.2 Master-Slave Solution 1

- Master and slave evaluate the brightness.
- Master switches LSAB if brightness is below its threshold.

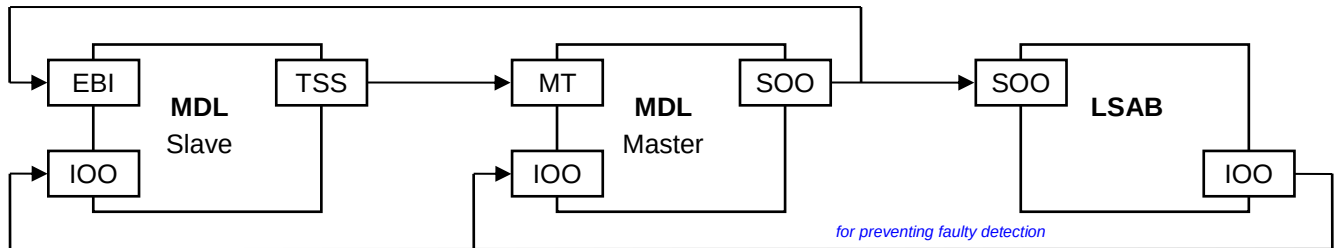


Figure 39 – Master-Slave Solution 1

USE CASE 7 It may happen that the FB MDL Master detects the first movement and switches the FB LSAB. It may then however occur that FB MDL Master does not detect any further movement and would switch off FB LSAB after timer OCT in the FB MDL Master expires, even if FB MDL Slave detects movement. This happens if the person has moved out of the detection area of FB MDL Masters into the detection area of the FB MDL Slave. This requires that the FB MDL Slave operates independently of the light brightness. This is done by linking the Output SOO of the FB MDL Master with the Input EBI of the FB MDL Slave.

USE CASE 8 If the FB MDL Slave detects the first movement and the brightness is below the threshold, it will trigger the FB MDL Master, which shall take the final conclusion about switching the light, depending on its further parameters. If the person moves from the detection area of the FB MDL Slave into the detection area of the FB MDL Master, then the FB MDL Master detects its first movement: the OCT in the FB MDL Master is retriggered by the own movement detection in the FB MDL Master and no longer from its Input MT. This is independent of the brightness detected by the FB MDL Master: with a running OCT, the brightness is not evaluated (see 2.2.5).

2.2.10.3 Master-Slave Solution 2

- Master and slave evaluate the brightness.

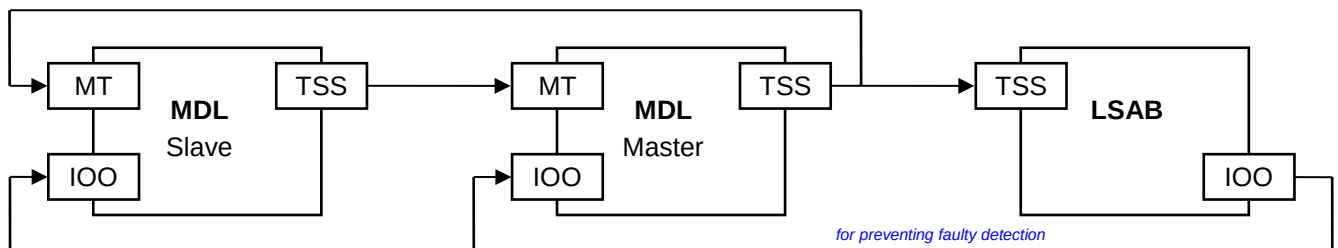


Figure 40 – Master-Slave Solution 2

USE CASE 9 The FB MDL Master detects the first movement and switches the FB LSAB through its Output TSS. If the person moves from the detection area of the FB MDL Master into the detection area of the FB MDL Slave then the FB MDL Master will not detect any further movement and the LSAB switches off the light autonomously after its switch off delay has expired, even if FB MDL Slave detects movement. This requires that the FB MDL Slave operates independently of the light brightness. This is done by linking the Output TSS of the FB MDL Master with the Input MT of the FB MDL Slave. This shall cause the timer MSLT in the FB MDL Slave to be started, which shall in turn cause that the FB MDL Slave operates brightness independently. However, it shall be avoided that the FB MDL Slave transmits on its Output TSS back to MT of the FB MDL Master: this will cause as many transmissions of TSS signals on the bus as there are FB MDL Slaves connected to the master. This can be avoided by appropriate setting of the Parameter Movement Trigger Evaluation.

USE CASE 10 If the FB MDL Slave detects the first movement and the brightness is below the threshold, it will trigger the FB MDL Master, which shall take the final conclusion about switching the light, depending on its further parameters. If the person moves from the detection area of the FB MDL Slave into the detection area of the FB MDL Master, then the FB MDL Master detects its first movement: the OCT in the FB MDL Master is retriggered by the own movement detection in the FB MDL Master and no longer from its Input MT. This is independent of the brightness detected by the FB MDL Master: with a running OCT, the brightness is not evaluated (see 2.2.5).

2.2.11 Extended functionality

The functionality of this FB Movement Detector for Lighting in an implementation can be extended by combining functionality of other FBs.

EXAMPLE 8 In the below example of Figure 41, FB MDL is combined with FB Scheduler. This may allow for instance to dim lights up or down to different values when movement is detected or no longer detected, to control shutters and blinds and much other functionality.

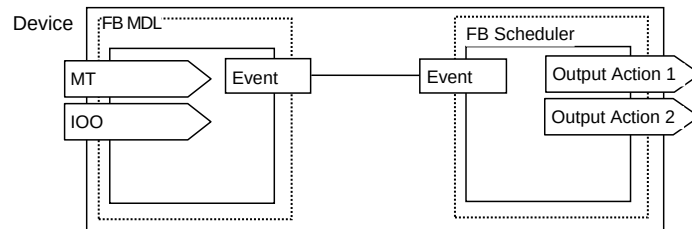


Figure 41 – FB MDL combined with FB Scheduler

2.3 Constraints

For evaluation of the illuminance from an external illuminance sensor, this FB MDL solely provides the Input Brightness External (BE), which may be compared to internal threshold values or an internal illuminance sensor. If the external illuminance sensor has an own evaluation of the illuminance and provides a 1 bit Output, then this can be linked to the Input Enable.

2.4 Functional Block diagram

FB Movement Detector for Lighting (MDL)			414
Inputs		Outputs	
Info OnOff	(IOO)	(SOO) M1	Switch OnOff
Enable	(EN)	(TSS) M1	Timed StartStop
Brightness External	(BE)		
Movement Trigger	(MT)		
additional I/Os		Parameters	
illuminance sensor (threshold)		(BT)	Brightness Threshold
PIR sensor		(OCT)	Output Control Time
		(TC)	Trigger Continue
		(EBI)	Enable Brightness Independency
		(MSPT)	Movement Sensor Pause Time
		(MTE)	Movement Trigger Evaluation

M1 Either one of these Datapoints is mandatory. See 2.6.

mandatory optional

Figure 42 – Functional Block Diagram for FB Switching Sensor Basic

2.5 Datapoints

Datapoint	Description/Remarks	Datapoint Type
Outputs		
Switch OnOff	Switches on or off the light.	1.001 DPT_Switch
Timed StartStop	Starts and retriggers the autonomous switching functionality at the linked actuators.	1.010 DPT_Start
Inputs		
Info OnOff	Informs on the light state of the actuator to suppress faulty detection of movement.	1.001 DPT_Switch

Datapoint	Description/Remarks	Datapoint Type
Enable	Enables and disables the detection of movement.	1.003 DPT_Enable
Brightness External	Receives the external brightness level.	9.004 DPT_Value_Lux
Movement Trigger	Add an external physical movement detector signal to the FB.	1.010 DPT_Start
Parameters		
Brightness Threshold	Threshold value above which the detection of movement is disabled.	9.004 DPT_Value_Lux
Output Control Time	The time after that the movement detector will wait for switching off the light after the last detection of movement (expiration of MSLT timer) via SOO, or the time in-between transmissions of TSS if TSS is repeated.	7.008 DPT_TimePeriodSec
Trigger Continue	Defines the dependency of the FB MDL from its Input IOO.	1.003 DPT_Enable
Enable Brightness Independency	Controls whether or not the functionality depends on the received or measured brightness.	1.003 DPT_Enable
Movement Sensor Pause Time	Short period after reception of the value on the Input IOO during which the detection of movement shall be suppressed.	7.004 DPT_TimePeriod100Msec
Movement Trigger Evaluation	Specifies the handling of the Input Movement Trigger.	1.003 DPT_Enable

2.6 FB Profiles⁶⁾

- FB Profile 1 matches use case 1: the switch off delay of an actuator is handled in the movement detector
- FB Profile 2 matches use case 2: the switch off delay of an actuator is handled in the actuator itself

Features and options	Basic FB	Standard Mode	
		FB Profile 1	FB Profile 2
select 1 of 2 {			
Output SOO + transmission and repetition conditions	M	GO	(GO)
Output TSS + transmission and repetition conditions	M	(GO)	GO
}			
Physical Movement Sensor	M	M	M
Inhibit false detections {	M	M	M
Input IOO	O	(GO)	(GO)
select 1 of 2 {	M	M	M
fixed pause time	M	M	M
Parameter MSPT	M	M	M
}			
Timer MSLT and functionality	M	M	M
Timer OCT and functionality	M	M	M
Parameter OCT time-out value	M	M	M

⁶⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

Features and options	Basic FB	Standard Mode	
		FB Profile 1	FB Profile 2
IF Parameter EBI is implemented {			
brightness dependency according EBI	M	M	M
}			
ELSE {			
FB MDL shall work brightness dependent	M	M	M
}			

2.7 Detailed specification of the Datapoints

2.7.1 Output Switch OnOff (SOO)

DP Name:	Switch OnOff	Abbr.:	SOO	Mandatory	☐
FB Name:	FB Movement Detector for Lighting (MDL)			Can be internal	☐
Description					
Please refer to the functional specification. This Output shall serve for switching on and off the light. This Output shall transmit the value On on the first detection of movement. It shall transmit the value Off after the period OCT (see 2.7.7) after the last detection of movement.					
Datapoint Type					
DPT_Name:	DPT_Switch				
DPT Format:	B ₁	DPT_ID:			
Field		Supp.	Range	Unit	Default
b	See DPT definition.	M	{0, 1}	none	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:					
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

2.7.2 Output Time StartStop (TSS)

DP Name:	Timed StartStop		Abbr.:	TSS	Mandatory	☐															
FB Name:	FB Movement Detector for Lighting (MDL)			Can be internal	☐																
Description																					
Please refer to the functional specification. This Output shall service for switching on the light. This can be used with a lighting actuator that switches off the light autonomously, e.g. after a certain period. This Output value shall be encoded according DPT_Start (1.010). Only the value 1 = start shall be transmitted; the value 0 = stop shall not be transmitted. As long as the FB MDL detects movement it shall send the value 1 = start periodically with the period according the Parameter OCT.																					
Datapoint Type																					
DPT_Name:	DPT_Start																				
DPT Format:	B ₁		DPT_ID:	1.010																	
Field			Supp.	Range	Unit	Default															
b	This shall switch on the light.		M	{0}	none	0															
Access Type																					
◆ Output																					
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☒</td> </tr> <tr> <td>Request</td> <td>☒</td> <td>Period:</td> <td>See 2.7.7</td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☐			Cyclic	☒	Request	☒	Period:	See 2.7.7	Δ-Value:			Min repetition time:	
this → M	☒	this → 1	☐																		
Spontaneous	☒	COV:	☐																		
		Cyclic	☒																		
Request	☒	Period:	See 2.7.7																		
Communication Type																					
◆ Group Object Datapoint					Mandatory:	☒															
Default Group Address:																					
Dynamics																					
Power down:	Save:	☐																			
Power up:	Value:	No initialisation:	☒	Default value:	☐																
		Saved value:	☐	Current value (not for input):	☐																
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																
Exception Handling																					
None.																					
Special Features																					
None.																					

2.7.3 Input Info OnOff (IOO)

DP Name:	Info OnOff	Abbr.:	IOO	Mandatory	☐
FB Name:	FB Movement Detector for Lighting (MDL)			Can be internal	☐
Description					
Please refer to the functional specification. When the light switches on or off in the spatial area in which the FB Movement Detector is physically mounted, this may by the electronics erroneously be detected as a movement. To prevent from this, the information to switch the light (e.g. from the light sensor) can be linked also to this Input Info OnOff. If the FB MDL receives a telegram on this Input, it will for a short period, possibly according the parameter Parameter Movement Sensor Pause Time2.7.11, not evaluate any movement sensor signal (internal signals MSLT).					
Datapoint Type					
DPT_Name:	DPT_Switch				
DPT Format:	B ₁	DPT_ID:			
Field		Supp.	Range	Unit	Default
b	0: The light is or will be switched off. 1: The light is or will be switched on.	M	{0, 1}	none	none
Access Type					
◆ Input					
N → this	☐	1 → this	☒		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:					
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

2.7.4 Input Enable (EN)

DP Name:	Enable	Abbr.:	EN	Mandatory	☐
FB Name:	FB Movement Detector for Lighting			Can be internal	☐
Description					
Please refer to the functional specification. This Input shall serve to limit the transmission of the Outputs SOO and/or TSS. The transmissions may either be disabled totally, or be limited to transmissions only caused by the Input MT.					
Datapoint Type					
DPT Name:	DPT_Enable				
DPT Format:	B ₁	DPT ID:	1.003		
Field		Supp.	Range	Unit	Default
b	0 = disable The transmissions of the Outputs SOO and/or TSS are limited. The movement signals that are caused by the built-in movement detector shall be disabled. Other conditions for the transmission of SOO and/or TSS may still be evaluated ^{a)} . 1 = enable The transmissions of the Outputs SOO and/or TSS shall not be limited by this Input.	M	{0, 1}	none	enable
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:					
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☐	Default value:	☒
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
^{a)} It may happen that the FB MDL is disabled while the timer OCT is running. When OCT expires, it may thus still happen that the FB MDL switches off the light. This is implementation dependent.					
Special Features					
None.					

2.7.5 Input Brightness External (BE)

DP Name:	Brightness External	Abbr.:	BE	Mandatory	☐
FB Name:	Movement Detector for Lighting			Can be internal	☐
Description					
This Input shall allow for the reception of the current brightness provided by an external room light sensor. This allows – for a FB MDL without built-in brightness sensor, or – for extra brightness information in addition to the brightness measured by the built-in brightness sensor.					
Datapoint Type					
DPT Name:	DPT_Value_Lux				
DPT Format:	F ₁₆	DPT ID:	9.004		
Field		Supp.	Range	Unit	Default
FloatValue	This field shall be the current illumination from the external brightness sensor.	M		Lux	none
Access Type					
◆ Input					
N → this	☐	1 → this	☒		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:					
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

2.7.6 Input Movement Trigger (MT)

DP Name:	Movement Trigger	Abbr.:	MT	Mandatory	☐
FB Name:	Movement Detector for Lighting			Can be internal	☐
Description					
This Input shall allow for the reception of external movement signals. It can be used as an alternative to the internal signal MSLT or be evaluated in parallel with it. The precise point where this signal is evaluated in the FB MSLT design is not fixed: it may or may not be subject to evaluation of IOO, brightness, timers, etc. The functionality of the Input MT can be disabled through the Parameter Movement Trigger Evaluation (see 2.7.10).					
Datapoint Type					
DPT Name:	DPT_Start				
DPT Format:	B ₁	DPT ID:	1.010		
Field		Supp.	Range	Unit	Default
b	0 = Stop This shall be interpreted as the end of the movement detection by an external movement sensor. 1 = Start This shall be interpreted as the start of the movement detection by an external movement sensor.	M	{0, 1}	none	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:					
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
		Saved value:	☐	Current value (not for input):	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	
Exception Handling					
None.					
Special Features					
None.					

2.7.7 Parameter Brightness Threshold (BT)

DP Name:	Brightness Threshold	Abbr.:	BT	Mandatory	☐
FB Name:	Movement Detector for Lighting (MDL)			Can be internal	☒
Description					
This Parameter shall be the threshold value of the detected brightness above which the movement detector will not signal detected movement, as specified in 2.2.5. This Parameter may be implemented as well as Memory Mapped Datapoint, as Group Object or Interface Object Property. The functionality can also be realised by a hardwired input.					
Datapoint Type					
DPT_Name:	DPT_Value_Lux				
DPT Format:	F ₁₆	DPT_ID:	9.004		
Field		Supp.	Range	Unit	Default
F ₁₆	Threshold value	M	cs	Lux	none
Access Type					
◆ Input					
N → this	☐	1 → this	☒		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☐
Default Group Address:		none			
◆ Property				Mandatory:	☐
DP Address: (in the server)	IO Type(ID):	414	Property ID:	51	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	3	Write level	3	
Dynamics					
Power down:	Save:	☒ ¹⁾			
Power up:	Value:	No initialisation:	☐	Default value:	☒
		Saved value:	☒	Current value (not for input):	☐
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
¹⁾ On power up, either the stored value may be used or the value ex-factory. If the ex-factory value is used, it is not required that this Parameter value be stored at power down.					
Special Features					
None.					

2.7.8 Parameter Output Control Time (OCT)

FB:	MDL	Property Name (Server):	Output Control Time			Mandatory	☐
						Optional	☒
Description:							
Please refer to the functional specification.							
For the Output Timed StartStop							
The Parameter OCT shall in this case be the period of the periodic transmission of the Output TSS in case movement is detected continuously for a longer time; see Figure 35.							
For the Output Switch OnOff							
The Parameter OCT shall in this case be the period starting with the last detection of movement and after which the FB MDL shall switch off the light; see Figure 34.							
DPT:	Name	DPT TimePeriodSec	DPT_ID	7.008	Datatype format:	U ₁₆	
Field	Description		Sup.	Range	Unit	Resol.:	Default
b	Time period.		M	min.: 0 s max: none	s	1 s	none
Communication:							
DP Address: (in the server)		Object Type:	414	Property ID:		110	
		Start-Index:	1	N° of elements			
Property access:		Read only ☐	-	Read/Write ☒			
Protection		Read level	-	Write level	-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
None.							
Special Features:							
None.							

2.7.9 Parameter Enable Brightness Independency (EBI)

DP Name:	Enable Brightness Independency	Abbr.:	EBI	Mandatory	☐
FB Name:	Movement Detector for Lighting			Can be internal	☐
Description					
This Parameter shall control (enable/disable) whether the FB MDL – functions independent of the measured or received brightness (Input Brightness External, see 2.7.5), and will thus switch the light actuator even if there is sufficient light in the room, or – functions only if the measured or received brightness is below a given threshold. This Parameter shall be implemented as a Group Object.					
Datapoint Type					
DPT Name:	DPT_Enable				
DPT Format:	B ₁		DPT ID:	1.003	
Field		Supp.	Range	Unit	Default
b	0 = disable (brightness is evaluated) The FB MDL shall only evaluate and possibly transmit detected movement if the brightness is below the threshold. 1 = enable (brightness is not evaluated). The FB MDL shall evaluate any detected movement regardless of the current brightness.	M	{0, 1}	none	1 = enable
Access Type					
◆ Input					
	N → this	☐	1 → this	☒	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:					
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☐	Default value:
		Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

2.7.10 Parameter Movement Trigger Evaluation

FB:	MDL	Property Name (Server):	Movement Trigger Evaluation		Mandatory	☐
					Optional	☒
Description:						
This Parameter shall control whether the Input Movement Trigger directly controls the Outputs SOO and/or TSS, or is subject to one or more internal evaluation criteria, just as MSLT state, and may thus not cause a transmission of SOO and/or TSS.						
DPT:	Name		DPT_ID		Datatype format:	
Field	Description		Sup.	Range	Unit	Resol.: Default
b	0 = disable The Input Movement Trigger does not directly affect the transmission of the Outputs SOO and/or TSS, but is subject to one or more internal evaluations. 1 = enable The Input Movement Trigger shall directly cause the transmission of the Outputs SOO and/or TSS.		M	{0,1}	none	none none
Communication:						
DP Address: (in the server)		Object Type:	414	Property ID:		111
		Start-Index:	1	N° of elements		
Property access:		Read only ☐	-	Read/Write ☒		
Protection		Read level	-	Write level		-
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐						
None.						
Special Features:						
None.						

2.7.11 Parameter Movement Sensor Pause Time

FB:	MDL	Property Name (Server):	Movement Sensor Pause Time		Mandatory	☐
					Optional	☒
Description:						
Please refer to the functional specification.						
This Parameter shall contain the short period starting with the reception of the value On on the Input IOO, during which the internal detection of light shall be suppressed.						
DPT:	Name	DPT_TimePeriod100Msec	DPT_ID	7.004	Datatype format:	U ₁₆
Field	Description		Sup.	Range	Unit	Resol.: Default
TimePeriod	This field shall contain the duration of the pause time.		M	min.: 0 s max: none	s	100 ms none
Communication:						
DP Address: (in the server)		Object Type:	414	Property ID:		112
		Start-Index:	1	N° of elements		
Property access:		Read only ☐	-	Read/Write ☒		
Protection		Read level	-	Write level		-
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐						
None.						
Special Features:						
None.						

2.7.12 Parameter Trigger Continue

FB:		Property Name (Server):	Trigger Continue (TC)		Mandatory	☐	
					Optional	☒	
Description:							
This Parameter shall allow that the FB MDL only controls the light while it knows via its Input IOO that the light is already switched on. The FB MDL will then keep the light on as long as it detects movement, this is, as long as none of its timers expire. This allows the realisation of USE CASE 6.							
DPT:	Name	DPT_Enable	DPT_ID	1.003	Datatype format:	B ₁	
Field	Description		Sup.	Range	Unit	Resol.:	Default
b	0 = disable The motion detector shall function independently of the value of the Input IOO, except for the suppression of the switching of the light. 1 = enable The FB MDL shall only evaluate the internal signal <i>MSLT state</i> if the Input IOO has the value "on". If the Input IOO has the value "off" then the internal signal <i>MSLT state</i> shall not be evaluated. Other timers and functions in the FB MDL shall be evaluated without change.		M	{0,1}	none	n/a	none
Communication:							
DP Address: (in the server)		Object Type:	414	Property ID:		113	
		Start-Index:	1	N° of elements			
Property access:		Read only	☐	-	Read/Write	☒	
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
None.							

3 Photometric sensors

3.1 Overview photometric sensors (informative)

The following types of photometric sensors exist.

1. Indoor Brightness Sensor This measures the indoor brightness.
This is modelled by the FB Indoor Brightness Sensor (FB IBS, 409) in this paper.
Its main Output provides the illuminance (DPT_Value_Lux, 9.004).
2. Indoor Luminance Sensor This measures the indoor luminance.
This is modelled by the FB Indoor Luminance Sensor (FB ILS, 410) in 3.2.
Its main Output provides the luminance (DPT_Value_Luminance, 14.041).
3. Daylight sensor This measures the outdoor brightness, typically in a larger range, for instance 1 000 Lux to 100 000 Lux.
It provides the illuminance to the bus and may foresee some 1 bit Outputs for threshold functions.
This sensor is not yet modelled as FB in KNX.
4. Twilight sensor This measures the outdoor brightness, typically in the lower range 0 Lux to 1 000 Lux.
It mainly provides a 1 bit Output to the bus.
This sensor is not yet modelled as FB in KNX.
5. Sun Intensity Sensor This sensor measures the sun intensity and provides the data to the bus with unit Wm^{-2} . This information is mainly used in HVAC applications.
This sensor is modelled in [03].

3.2 FB Indoor Brightness Sensor (Object Type 409)

3.2.1 Aims and objectives

The FB Indoor Brightness Sensor (FB IBS) shall provide the illuminance measured at a certain location and provide this to the KNX system.

NOTE 5 This relates in turn to the irradiance (Wm^{-2}) as specified in FB Sun Intensity Sensor (see [03]) weighted by the spectral sensitivity of the human eye.

This shall be a physical sensor with a built-in or hard-wired brightness sensor.

3.2.2 Functional specification

3.2.2.1 Explanation of illuminance

The illuminance is the ratio of the luminous flux incident on a given surface and the size of that surface.

Typical illuminance value that can be expected are listed in Table 15.

Table 15 – Typical illuminance values

Light conditions	Illuminance
Sunlight at noon in summer	100 000 Lux
Cloudy sky in summer	10 000 Lux
Rainy wheather with thunder clouds	1 000 Lux
Office lighting	500 Lux
Living room lighting	200 Lux
Staircase lighting	100 Lux
Street lighting	10 Lux
Twilight after sunset	1 Lux
Midnight at full moon	0,2 Lux
Sky with stars and no moon	0,000 5 Lux

3.2.2.2 Basic functionality

The FB IBS shall be able to detect the indoor brightness.

From the measured electronic signals, the illumination shall be calculated. This calculation is not standardised. It shall lead to the value representing the illumination on a given room element. This calculated value shall be transmitted to KNX via the Output Room Illumination encoded as DPT_Value_Lux (DPT_ID: 9.004).

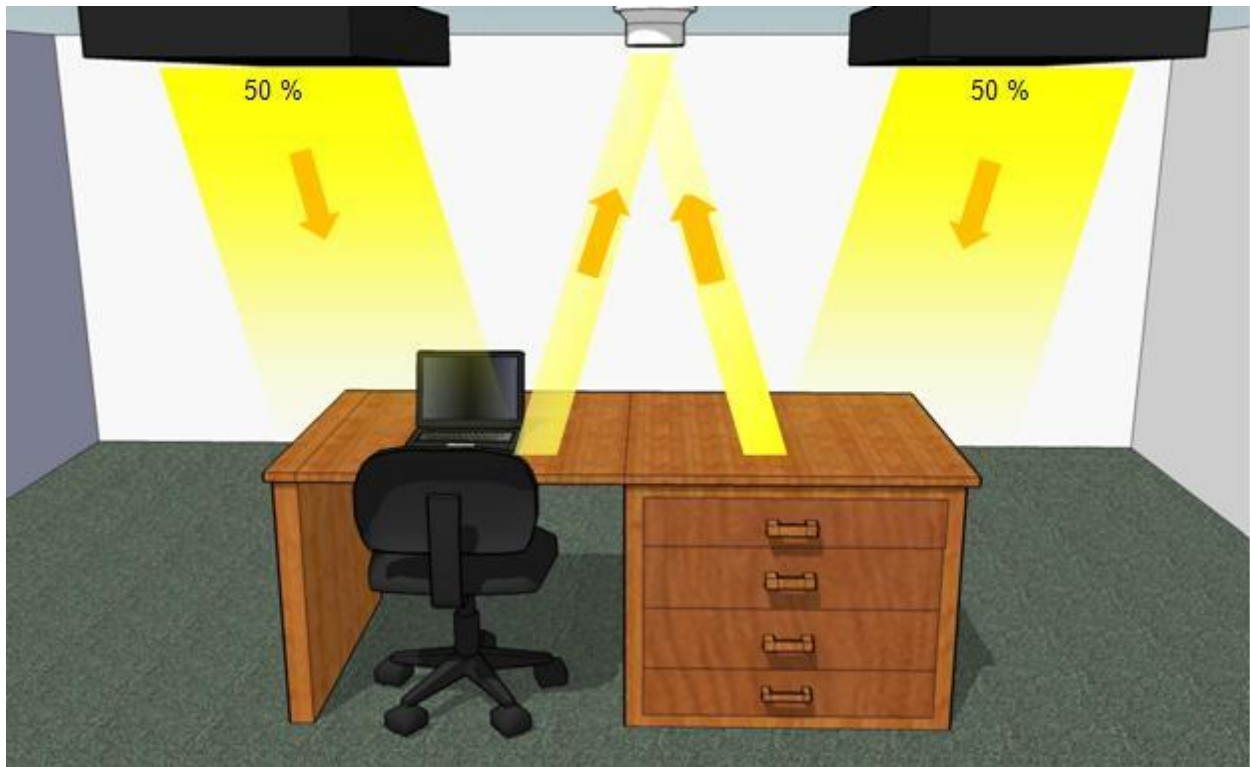


Figure 43 – Typical room brightness measurement method

3.2.2.3 Calibration

It may be necessary that the installed device that realises the FB Brightness Sensor has to be calibrated. This may be due to:

- the electronic measurement method, or
- the mechanics (optics), or
- the mounting place (e.g. in the ceiling), or
- the reflection by objects in the room (e.g. floor, furniture, walls, etc.)

The installer procedure for this calibration is not standardised.

3.2.3 Constraints

3.2.3.1 Evaluation of the calculated illuminance

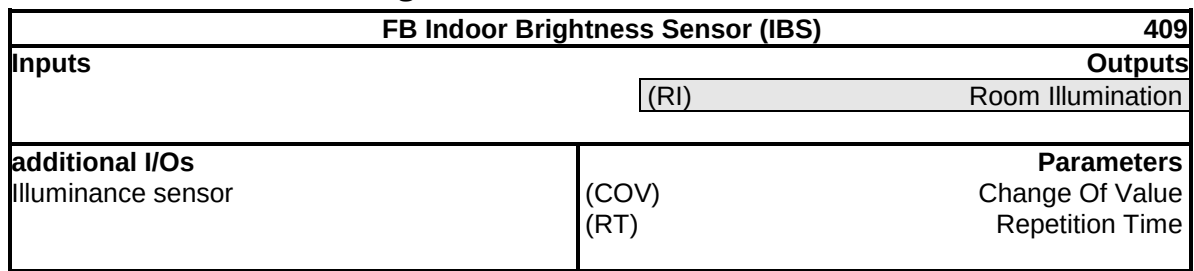
The FB IBS does not further evaluate the illuminance. The evaluation of threshold values or the further control of lighting is the task for different Functional Blocks.

3.2.3.2 Luminance (cdm^{-2})

This FB IBS does not feature any measurement of the luminance. The luminance shall be provided by the FB Indoor Luminance Sensor (see [03]).

If it is wanted that both illuminance as well as luminance be provided on the bus, then both FBs Indoor Brightness Sensor and Indoor Luminance Sensor can be combined in a device.

3.2.4 Functional Block diagram



 mandatory  optional

Figure 44 – Functional Block Diagram for FB Indoor Brightness Sensor

3.2.5 Datapoints

Datapoint	Abbr.	Description	Datapoint Type
Outputs			
Room Illumination	RI	Provides the calculated illuminance to the bus.	9.004 DPT_Value_Lux
Parameters			
Change Of Value	COV	Change of value in lux	9.004 DPT_Value_Lux
Repetition Time	RT	Repetition time value in seconds	7.005 DPT_TimePeriodSec

3.2.6 FB Profiles ⁷⁾

Features and options	Basic FB	Standard Mode
		IBS FB Profile 1
Output RoomIllumination {		
Output RoomIllumination	M	GO
Change of Value	M	M
Parameter COVLux	O	O
Parameter COVPercent	O	O
Periodic transmission	M	M
}		

⁷⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

3.2.7 Detailed specification of the Datapoints

3.2.7.1 Output Room Illumination (RI)

DP Name:	Room Illumination	Abbr.:	RI	Mandatory	☒
FB Name:	FB Indoor Brightness Sensor			Can be internal	☐
Description					
This Datapoint shall represent the illumination calculated by the FB. If the variation of calculated illumination changes more than COV then the newly calculated value shall be transmitted on the bus. If the illumination doesn't change more than COV, then the current calculated value shall be transmitted periodically.					
Datapoint Type					
DPT Name:	DPT Value Lux				
DPT Format:	F ₁₆	DPT ID:	9.004		
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value	See COV
		Cyclic	☒	Period:	See RT
Request	☐				
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					---
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☐	Default value:	☐
		Saved value:	☐	Actual value (not for input):	☒
Transmit on bus (only for output):				☐	Read from bus (only for input):
Exception Handling					
None.					
Special Features					
None.					

3.2.7.2 Parameter Change of Value (COV)

FB:	IBS	Property Name (Server):	Change of Value	Mandatory	☐	
				Optional	☒	
Description:						
This represents the minimal change of the measurement that send a new value on the Output RI.						
DPT:	Name	DPT Value Lux	DPT ID	9.004	Datatype format:	
					F ₁₆	
Field	Description		Sup.	Range	Unit	Resol.:
FloatValue	This field shall contain the value over which the measured illuminance shall change before it is transmitted on the bus.		M	cs	Lux	1 Lux
						cs
Communication:						
◆ Property					Mandatory:	
DP Address:	Object Type:	409	Property ID:	110		
(in the server)	Start-Index:	1	N° of elements	1		
Property access:	Read only	☐	Read/Write	☒		
Protection	Read level	-	Write level	-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐						
Special Features:						
In channel code PART_COV_Lux is used.						

3.2.7.3 Parameter: Repetition Time (RT)

FB:	IBS	Property Name (Server):	Repetition Time				Mandatory Optional	☐ ☒	
Description:									
This represents the period of emission of the Ouptut Room Illumination (RI) when no change of value is detected.									
DPT:	Name	DPT_TimePeriodSec	DPT_ID	7.005	Datatype format:		U ₁₆		
Field		Description		Sup.	Range	Unit	Resol.:	Default	
TimePeriod		This shall specify the repetition time of the periodic transmission of the Output RI.		M	cs	s	1 s	cs	
Communication:									
DP Address: (in the server)		Object Type:	409	Property ID:		111			
		Start-Index:	1	N° of elements		1			
Property access:		Read only	☐	-	Read/Write	☒			
Protection		Read level	-	Write level		-			
Exception Handling:		Value after Power-up:	Stored Value	☒	Curr Value	☐	Default Value		☐
None.									
Special Features:									
In channel code PART Cycle Time is used									

3.3 FB Indoor Luminance Sensor (Object Type 410)

3.3.1 Aims and objectives

The FB Indoor Luminance Sensor (ILS) shall provide the luminance measured at a certain location and provides this to the KNX system.

3.3.2 Functional specification

3.3.2.1 Explanation of luminance

Luminance is defined in [06]. It represents the luminous intensity in a certain direction.

This quantity characterizes the impression of brightness, which an illuminated or a luminant surface effects in the human eye.

The luminance is independent from the distance to the observer; that means, that the impression of brightness does not change when the distance between observer and observed object changes.

Typical luminance value that can be expected are listed in Table 16.

Table 16 – Typical luminance values

Light source	Luminance
Sun at noon	1 600 000 000 cd/m ²
100 W clear light bulb	10 000 000 cd/m ²
Sun at sunset	5 000 000 cd/m ²
100 W opal light bulb	200 000 cd/m ²
Blue sky	10 000 cd/m ²
Candlelight	5 000 cd/m ²
Moonlight	2 500 cd/m ²
sky at night	0,001 cd/m ²

3.3.2.2 Basic functionality

The FB ILS shall be able to detect the indoor luminance.

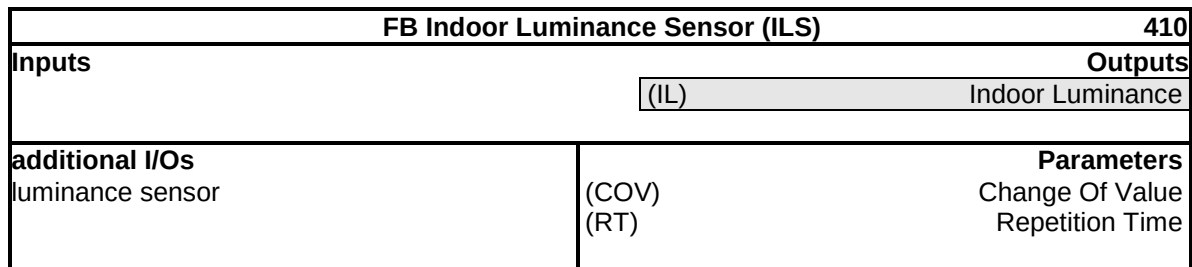
The luminance is typically measured by an electronic sensor.

The measured luminance shall be provided to the KNX system through the Output Indoor Luminance (IL) encoded as DPT_Value_Luminance (14.041, F₃₂, cdm⁻²).

3.3.3 Constraints

None.

3.3.4 Functional Block diagram



 mandatory  optional

Figure 45 – Functional Block Diagram for FB Indoor Luminance Sensor

3.3.5 Datapoints

Datapoint	Abbr.	Description	Datapoint Type
Outputs			
Indoor Luminance	IL	Provides the measured luminance to the bus.	14.041 DPT_Value_Luminance
Parameters			
Change Of Value	COV	Change of value in cdm ⁻² .	14.041 DPT_Value_Luminance
Repetition Time	RT	Repetition time value in seconds	7.005 DPT_TimePeriodSec

3.3.6 FB Profiles ⁸⁾

Features and options	Basic FB	Standard Mode
		ILS FB Profile 1
Output Indoor Luminance {		
Output IL	M	GO
Change of Value	M	M
Parameter COV_cd_per_m2	O	O
Parameter COVPercent	O	O
Periodic transmission	M	M
}		

3.3.7 Detailed specification of the Datapoints

3.3.7.1 Output Indoor Luminance (IL)

DP Name:	Indoor Luminance	Abbr.:	IL	Mandatory	☒			
FB Name:	FB Indoor Luminance Sensor			Can be internal	☐			
Description								
This Datapoint shall represent the luminance observed by the FB ILS. If the variation of the luminance changes more than COV then the new value shall be transmitted on the bus. If the luminance doesn't change more than COV, then the current value shall be transmitted periodically.								
Datapoint Type								
DPT_Name:	DPT_Value_Luminance							
DPT Format:	F ₃₂	DPT_ID:	14.041					
◆ Output								
	this → M	☒	this → 1	☐				
	Spontaneous	☒	COV:	☒	Δ-Value	See COV	Min repetition period:	none
			Cyclic	☒	Period:	See RT		
	Request	☐						
Communication Type								
◆ Group Object Datapoint				Mandatory:	☒			
Default Group Address:		---						
Dynamics								
	Power down:	Save:	☐					
	Power up:	Value:	No initialisation:	☐	Default value:	☐		
			Saved value:	☐	Actual value (not for input):	☒		
		Transmit on bus (only for output):		☐	Read from bus (only for input):		☐	
Exception Handling								
None.								
Special Features								
None.								

⁸⁾ Please refer to [02] for the definition of the syntax and symbols used in this FB Profile definition.

3.3.7.2 Parameter Change of Value (COV)

FB:	ILS	Property Name (Server):	Change of Value				Mandatory	☐
							Optional	☒
Description:								
This represents the minimal change of the measurement that send a new value on the Output IL.								
DPT:	Name	DPT_Value_Luminance	DPT_ID	14.041	Datatype format:	F ₃₂		
Field	Description		Sup.	Range	Unit	Resol.:	Default	
F ₃₂	This field shall contain the value over which the measured luminance shall change before it is transmitted on the bus.		M	cs	cdm ⁻²	1 cdm ⁻²	cs	
Communication:								
◆ Property						Mandatory:	☐	
DP Address: (in the server)		Object Type:	410	Property ID:		110		
		Start-Index:	1	N° of elements		1		
Property access:		Read only	☐	Read/Write	☒			
Protection		Read level	3	Write level	3			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐								
Special Features:								
None.								

3.3.7.3 Parameter: Repetition Time (RT)

FB:	ILS	Property Name (Server):	Repetition Time				Mandatory	☐
							Optional	☒
Description:								
This represents the period of emission of the Output Indoor Luminance (IL) when no change of value is detected.								
DPT:	Name	DPT_TimePeriodSec	DPT_ID	7.005	Datatype format:	U ₁₆		
Field	Description		Sup.	Range	Unit	Resol.:	Default	
TimePeriod	This shall specify the repetition time of the periodic transmission of the Output IL.		M	cs	s	1 s	cs	
Communication:								
DP Address: (in the server)		Object Type:	410	Property ID:		111		
		Start-Index:	1	N° of elements		1		
Property access:		Read only	☐	Read/Write	☒			
Protection		Read level	3	Write level	3			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐								
None.								
Special Features:								
None.								

4 Functional Blocks for E-Mode Channels

4.1 Usage requirements

The Functional Block specifications below only provide complementary information to the E-Mode Channel definitions. They are only provided for completeness and understanding of these E-Mode Channel definitions.

As a consequence, no Object Types for Interface Objects are specified below. It will be a decision of KNX Association Working Group Interworking how these Functional Blocks will be continued and if any Object Type will be assigned.

These Functional Blocks shall be used only for implementation of E-Mode devices.

These Functional Block specifications shall not be used for any other goal; in particular, no implementation for S-Mode devices shall be based on these specifications.

KNX Association will take care of compatibility between any currently specified E-Mode Channel definition and the final version of these Functional Blocks.

To this, the KNX Association Application Specification Groups shall take the functionality achieved by these Functional Blocks as the minimal mandatory basis for further work.

4.2 FB_Switch_Scene_Numbered

4.2.1 Definition

- **Name:** FB_Switch_Scene_Numbered
- **Object type:** See “Usage requirements” in 4.1.

4.2.2 Functional specification

On interaction, a scene number shall be sent.

There shall be two possible interactions.

1. On interaction 1 the value of parameter Scene Number 1 (P1) shall be sent.
2. On interaction 2 the value of parameter Scene Number 2 (P2) shall be sent.

It shall not be possible to learn scenes.

4.2.3 Functional Block diagram

FB_Switch_Scene_Numbered	
Inputs	Outputs
Parameters	(SN) Scene Number
Scene Number 1 (P1)	
Scene Number 2 (P2)	

4.2.4 Datapoints

Datapoint	Abbr.	Description	Datapoint Type
Outputs			
Scene Number	SN	To activate the scene of this value	18.001 DPT_SceneControl
Parameters			
Scene Number 1	P1	Scene number to send on interaction 1	18.001 DPT_SceneControl
Scene Number 2	P2	Scene number to send on interaction 2	18.001 DPT_SceneControl

4.2.5 Runtime Interworking – dependence on Configuration Mode

			Standard Mode	Extended Mode
		Basic FB	S-Mode	Standard Mode Interface
				LTE-HEE
Outputs	SN	GO	-	-
Parameters	P1		-	-
	P2		-	-

4.2.6 Detailed specification of the Datapoints

4.2.6.1 Output Scene number

DP Name:	Scene Number	Abbr.:	SN	Mandatory	☒
FB Name:	FB_Switch_Scene_Numbered			Can be internal	☐
Description					
This Datapoint shall hold the value that shall be sent on the bus on interaction.					
Datapoint Type					
DPT Name:	DPT_SceneControl				
DPT Format:	B ₁ R ₁ U ₆	DPT_ID:	18.001		
♦ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☐	Δ-Value:	Min repetition period:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
♦ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☐	Default value:	☐
		Saved value:	☐	Actual value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

4.2.6.2 Parameter Scene Number 1

DP Name:	Scene Number 1	Abbr.:	P1	Mandatory	☒
FB Name:	FB_Switch_Scene_Numbered			Can be internal	☐
Description					
This parameter shall be used to set the scene number sent on interaction 1 on the switch.					
Datapoint Type					
DPT Name:	DPT_SceneControl				
DPT Format:	B ₁ R ₁ U ₆	DPT_ID:	18.001		
Fields	Description	Supp.	Range	Unit	Default
B ₁	= 0 not used				
R ₁	Reserved				
U ₆	Scene number		[0 ... 63]		
Exception Handling					
None.					
Special Features					
None.					

4.2.6.3 Parameter Scene Number 2

DP Name:	Number 2	Abbr.:	P2	Mandatory	☒
FB Name:	FB_Switch_Scene_Numbered			Can be internal	☐
Description					
This parameter shall be used to set the scene number sent on interaction 2 on the switch.					
Datapoint Type					
DPT_Name:	DPT_SceneControl				
DPT Format:	B ₁ R ₁ U ₆			DPT_ID:	18.001
Fields	Description	Supp.	Range	Unit	Default
B ₁	= 0 not used				
R ₁	Reserved				
U ₆	Scene number		[0 ... 63]		
Exception Handling					
None.					
Special Features					
None.					

4.3 FB_Switch_Dimming_Value

4.3.1 Definitions

- **Name:** FB_Switch_Dimming_Value
- **Object type:** See “Usage requirements” in 4.1.

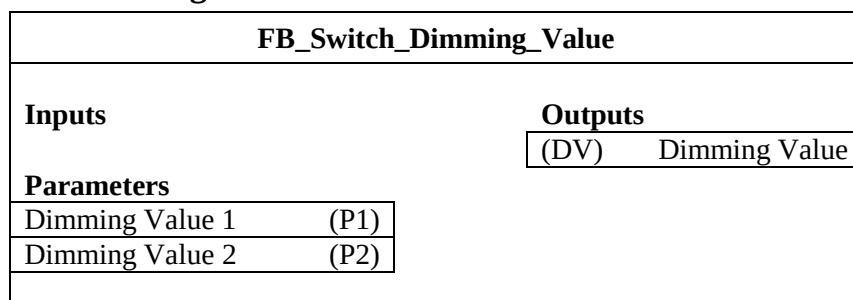
4.3.2 Functional specification

On interactions, a dimming value shall be sent.

There shall be two possible interactions.

1. On interaction 1 the value parameter Dimming Value 1 (P1) shall be sent.
2. On interaction 2 the value of parameter Dimming Value 2 (P2) shall be sent.

4.3.3 Functional Block diagram



4.3.4 Datapoints

Datapoint	Abbr.	Description	Datapoint Type
Outputs			
Dimming Value	DV	To set an absolute dimming value.	5.001 DPT_Scaling
Parameters			
Dimming Value 1	P1	Value of the dimming value to send on interaction 1.	5.001 DPT_Scaling
Dimming Value 2	P2	Value of the dimming value to send on interaction 2.	5.001 DPT_Scaling

4.3.5 Runtime Interworking – dependence on Configuration Mode

				Standard Mode		
					LTE	
		Basic FB	S-Mode	Standard Mode Interface	LTE-HEE	
Outputs	DV	GO		-	-	
Parameters	P1			-	-	
	P2			-	-	

4.3.6 Detailed specification of the Datapoints

4.3.6.1 Output Dimming Value

DP Name:	Dimming_Value	Abbr.:	DV	Mandatory	☒
FB Name:	FB_Switch_Dimming_Value			Can be internal	☐
Description					
This Datapoint shall represent the value that shall be sent on the bus on interaction.					
Datapoint Type					
DPT_Name:	DPT_Scaling				
DPT Format:	U ₈	DPT_ID:	5.001		
♦ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☐	Δ-Value:	Min repetition period:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
♦ Group Object Datapoint				Mandatory:	☒
Default Group Address:		---			
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☐	Default value:	☐
		Saved value:	☐	Actual value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
None.					
Special Features					
None.					

4.3.6.2 Parameter Value 1

DP Name:	Value 1	Abbr.:	P1	Mandatory	☒
FB Name:	FB_Switch_Dimming_Value			Can be internal	☐
Description					
This parameter is used to set the value sent on interaction 1 on the switch					
Datapoint Type					
DPT_Name:	DPT_Scaling				
DPT Format:	U ₈	DPT_ID:	5.001		
Exception Handling					
None.					
Special Features					
None.					

4.3.6.3 Parameter Value 2

DP Name:	Value 2	Abbr.:	P2	Mandatory	☒
FB Name:	FB_Switch_Dimming_Value			Can be internal	☐
Description					
This parameter is used to set the value sent on interaction 2 on the switch.					
Datapoint Type					
DPT_Name:	DPT_Scaling				
DPT Format:	U ₈	DPT_ID:	5.001		
Exception Handling					
None.					
Special Features					
None.					

4.4 FB_PB_Dimming_Value

4.4.1 Definition

- **Name:** FB_PB_Dimming_Value
- **Object type:** See “Usage requirements” in 4.1.

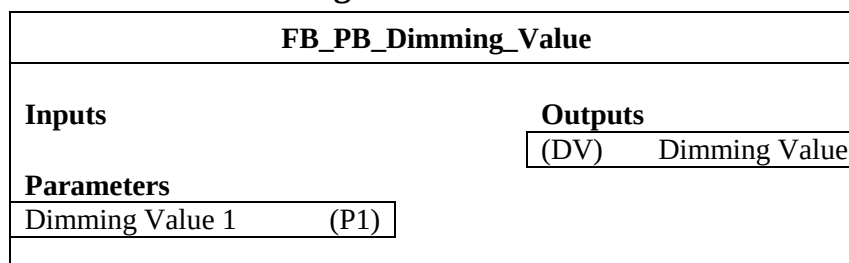
4.4.2 Functional specification

On interaction, a dimming value shall be sent.

There shall be only one possible interaction.

1. On interaction 1 (press on button for example) value of parameter Dimming Value (P1) shall be sent.

4.4.3 Functional Block diagram



4.4.4 Datapoints

Datapoint	Abbr.	Description	Datapoint Type
Outputs			
Dimming Value	DV	To set an absolute dimming value	5.001 DPT_Scaling
Parameters			
Dimming Value	P1	Value of the dimming value to send on interaction 1	5.001 DPT_Scaling

4.4.5 Runtime Interworking – dependence on Configuration Mode

				Standard Mode	Extended Mode
		Basic FB	S-Mode	Standard Mode Interface	LTE-HEE
Outputs	DV	GO		-	-
Parameter	P1			-	-

4.4.6 Detailed specification of the Datapoints

4.4.6.1 Output Dimming Value

DP Name:	Dimming_Value	Abbr.:	DV	Mandatory	☒
FB Name:	FB_PB_Dimming_Value	Can be internal	☐		
Description					
This Datapoint represents the value sent on the bus on interaction.					
Datapoint Type					
DPT_Name:	DPT_Scaling				
DPT Format:	U ₈	DPT_ID:	5.001		
♦ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☐	Δ-Value:	Min repetition period:
		Cyclic	☐	Period:	
Request	☐				
Communication Type					
♦ Group Object Datapoint				Mandatory:	☒
Default Group Address:	---				
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☐	Default value:	☐
		Saved value:	☐	Actual value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
None.					
Special Features					
None.					

4.4.6.2 Parameter Value

DP Name:	Value 1	Abbr.:	P1	Mandatory	☒
FB Name:	FB_PB_Dimming_Value			Can be internal	☐
Description					
This parameter is used to set the value sent on interaction 1 on the switch					
Datapoint Type					
DPT_Name:	DPT_Scaling				
DPT Format:	U ₈		DPT_ID:	5.001	
Exception Handling					
None.					
Special Features					
None.					